

Mortgage Credit Models & Projections

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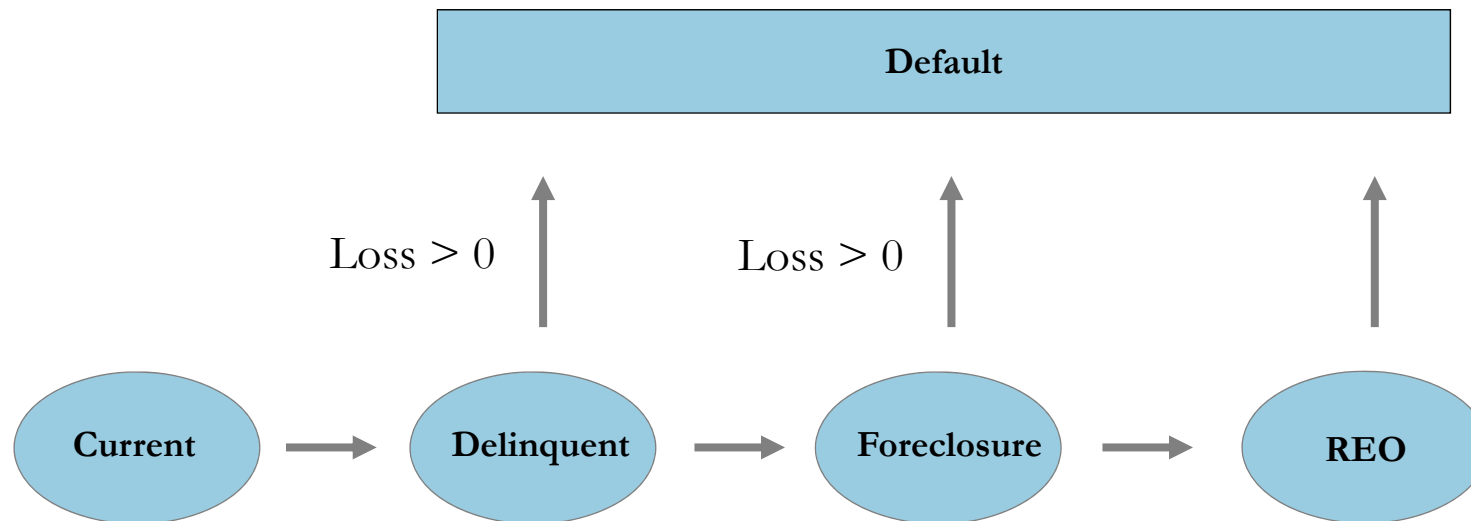
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8th May, 2014

Mortgage Valuation System - Components

- ***Voluntary Prepayment Model:*** projects voluntary speeds (VPR)
- ***Default Model:*** projects involuntary part (CDR) of the CPR
- ***Delinquency Roll Rate Model:*** transitions between states
- ***Loss Severity Model:*** % of remaining balance lost if loan defaults
- ***Term Structure Model:*** generates simulated yield curve paths
- ***MOATS Model:*** calculates mortgage rates as a function of yield curve and vols
- ***Home Price Simulation Model:*** projects home price changes as a function of past home prices and mortgage rates – needed for credit-adjusted OAS analysis

What is a Default ?



Source: Citi

We define default as a payoff (or prepayment) when the previous status was REO, or the previous status was delinquent or in foreclosure and there was a loss.

- **Delinquent:** when a borrower has missed one or more monthly payments.
- **Foreclosure:** is a legal process, the goal of which is to sell the collateral (the property) so that the proceeds can be used to meet contractual obligations of the mortgage. Takes 9 to 18 months depending on state laws.
- **REO:** At the end of the foreclosure process (or earlier), if the servicer takes possession of the property. Loans stay in REO an average of 6 months.

Integrated Prepayment & Default Models

$$\begin{aligned}\text{Prepayment Rate} &= (\text{Turnover} + \text{Refis} + \text{Curtailments}) + \text{Defaults} \\ &= \text{Voluntary Prepayments} + \text{Defaults}\end{aligned}$$

Default Component – function of

- **Loan Features:** Loan Type, WALA, Payment Shock, Downpayment, CLTV, Purpose, Lien Status, Documentation, SATO
- **Borrower Characteristics:** Credit Score (FICO), DTI, Occupancy Status
- **Macro-Economic Variables:** Home Price Appreciation, Home Price Momentum, Unemployment Rates, Interest Rates.

Interaction between **Voluntary** and **Involuntary Prepay** Components:

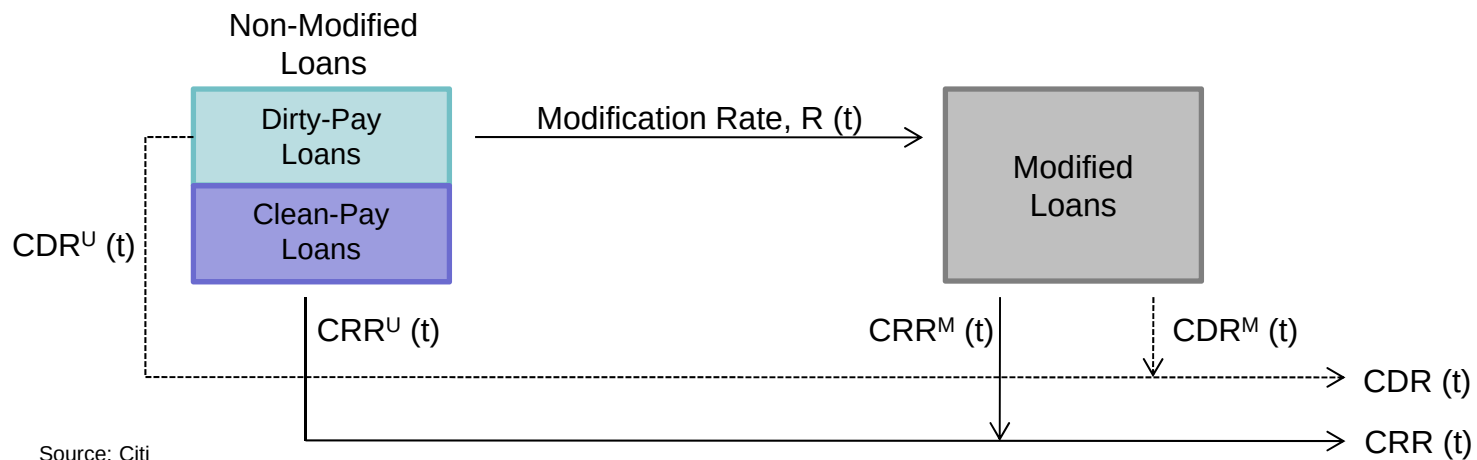
- **Refi Burnout** – better borrowers leave the pool
- **Credit Burnout** – worse borrowers leave the pool

Framework of the Default Model

Loan Modifications and Payment History Incorporated in Projections

- Collateral is divided into (a) **Modified** and (b) **Non-Modified** population
- Non-Modified population is further divided into **clean pay** and **dirty pay** borrowers based on their payment history.
- Modified population grows based on some mod rate ($R(t)$) assumption.
- Separate default model for each population.
- Long-term default and prepayment rate of modified loans converges to those that are not modified.

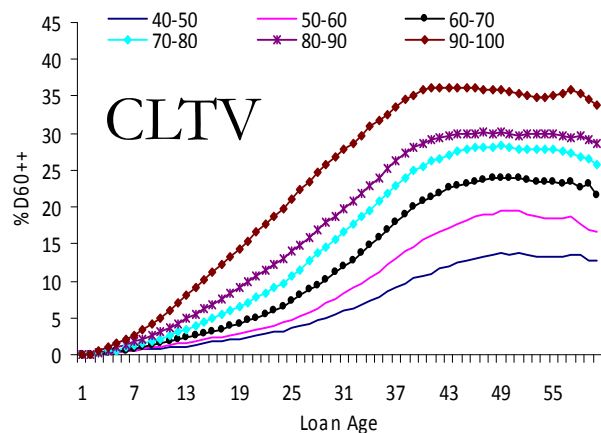
Framework for Modeling Effect of Loan Modifications on Prepays and Defaults



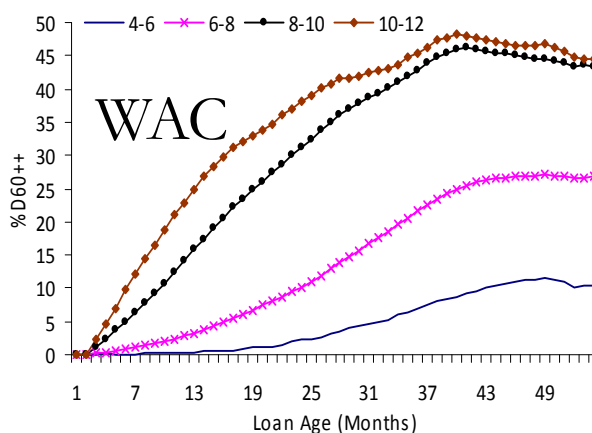
Loan and Borrower Characteristics Important For Default (Data: 2006 vintage)



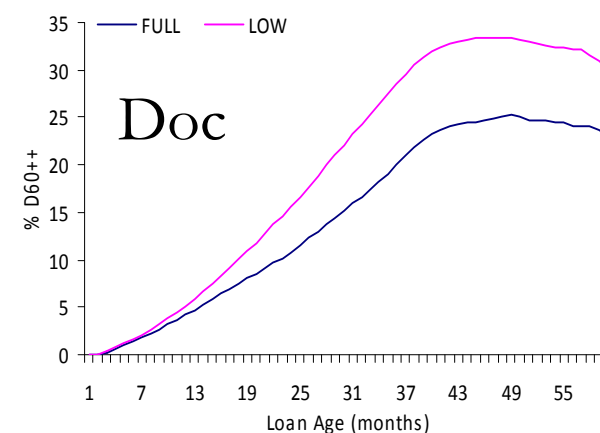
Loan Characteristics



Source: Citi, CoreLogic

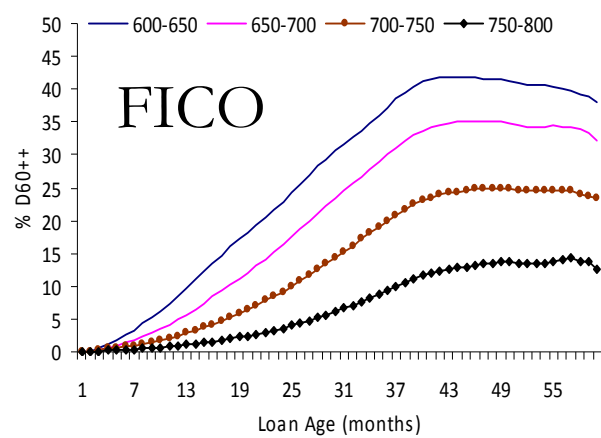


Source: Citi, CoreLogic

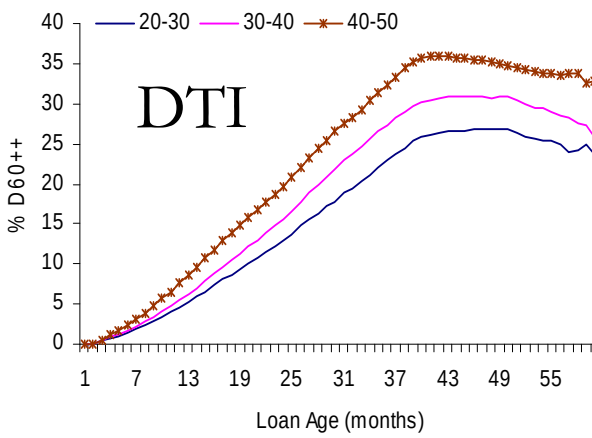


Source: Citi, CoreLogic

Borrower Characteristics



Source: Citi, CoreLogic



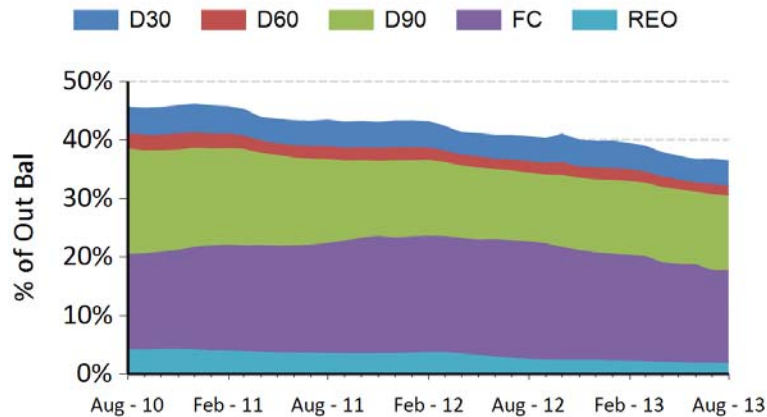
Source: Citi, CoreLogic

- ✓ **CLTV**
- ✓ **WAC**
- ✓ **Documentation**
- ✓ **FICO**
- ✓ **Debt-to-Income (DTI)**
- ✓ **Occupancy**
- ✓ **Loan Purpose**
- ✓ **Payment Shock**
- ✓ **Economic Factors**

See "Modeling of Mortgage Defaults," Citi, January 22, 2008.

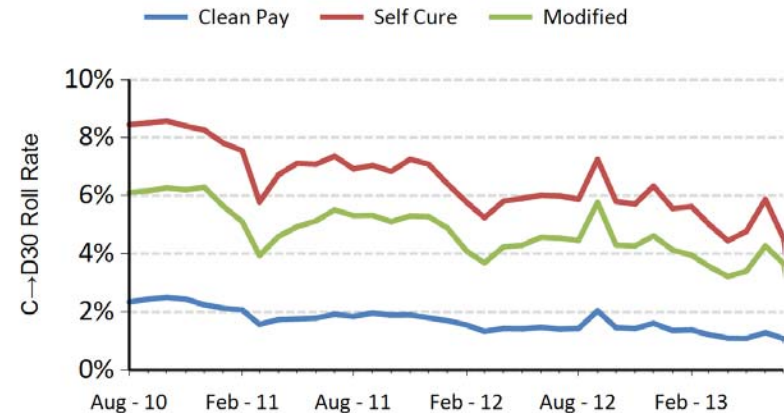
Delinquency Roll Rate Should Continue to Decline

Distribution by Delinquency Status (Subprime)



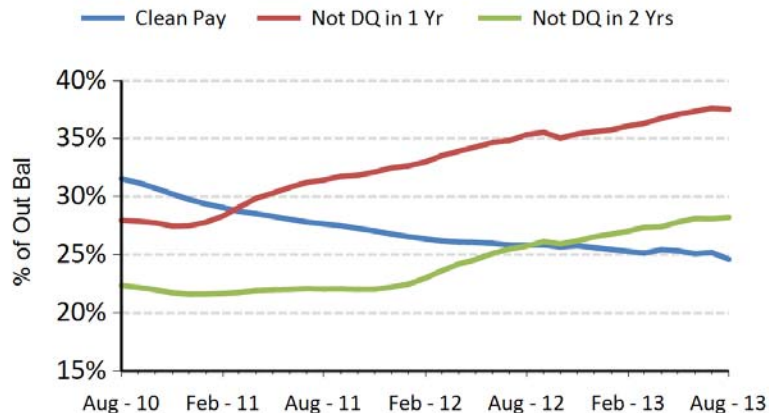
Source: Citi, CoreLogic

C→D30 RR by Pay History (Subprime)



Source: Citi, CoreLogic

Distribution by Payment History (Subprime)

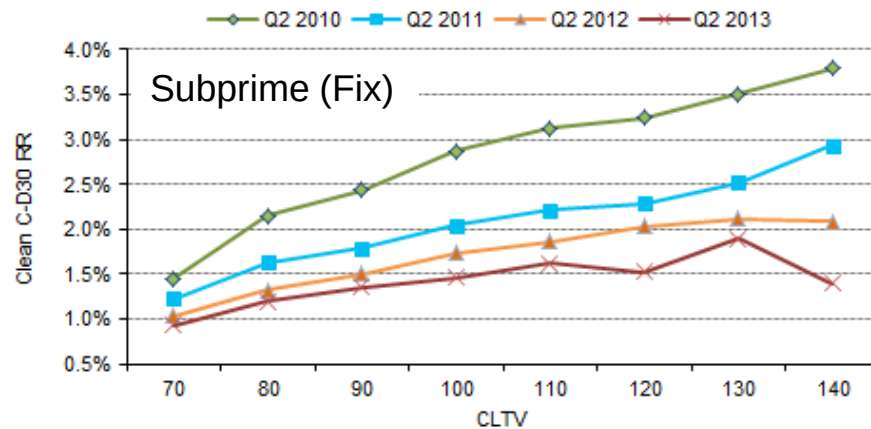


Source: Citi, CoreLogic

- Overall delinquencies have declined and will continue to do so due to lower C-D30 roll rates. Performance across all loans based on pay history have improved due to several reasons including positive HPA. Same factors should drive performance going forward although the impact may vary across different sectors.
- **Percentage of loans with more than 24 consecutive payments has been increasing consistently.**

Credit and Prepay Burnout – Expect Delinquencies to Continue to Decline

Subprime Continues to Experience Credit Burnout

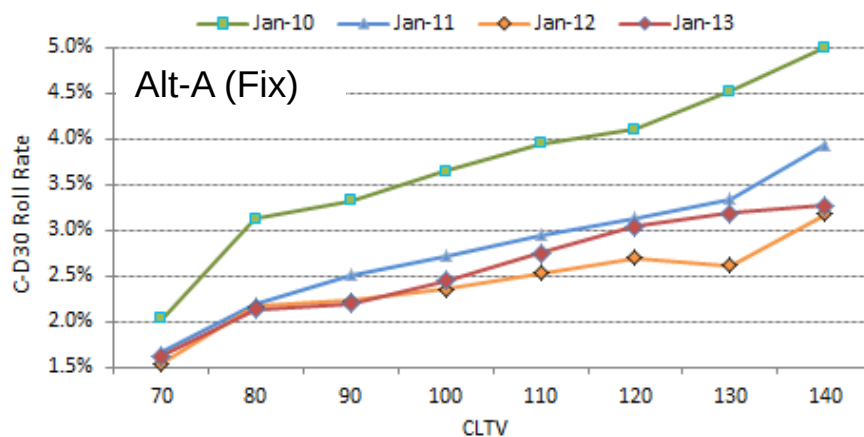


Source: Citi, CoreLogic

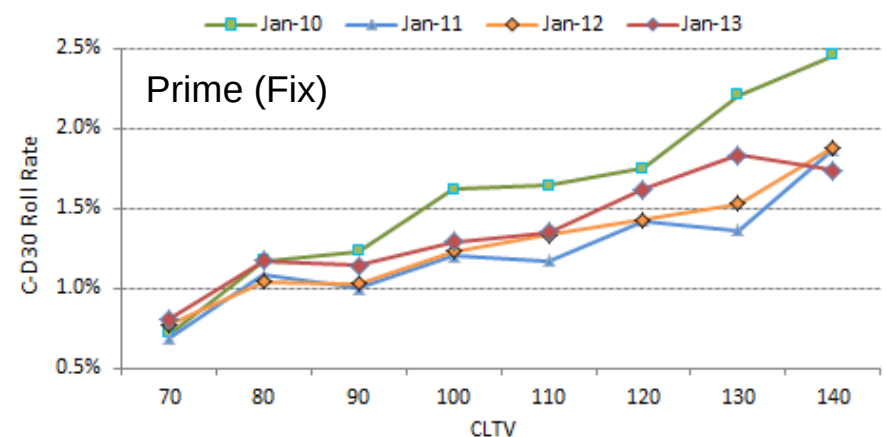
- Interaction between prepayment and default models.
- Credit-burnout: improving collateral as high defaults lead to worse borrowers leaving the pool.
- Prepay-burnout: deteriorating collateral as high voluntary prepayment leads to good borrowers leaving the pool.

Overall Delinquencies should continue to decline as there is a shift toward lower CLTV due to positive HPA.

Prime and Alt-A Fix Recent Roll Rates Are Higher Than a Year Ago Due to Prepay Burnout



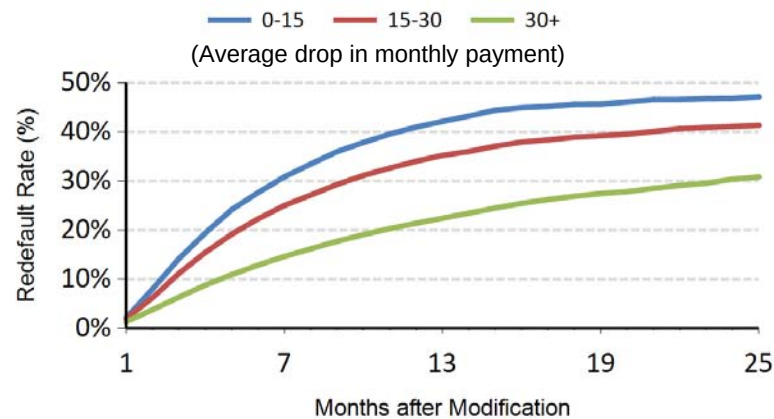
Source: Citi, CoreLogic



Source: Citi, CoreLogic

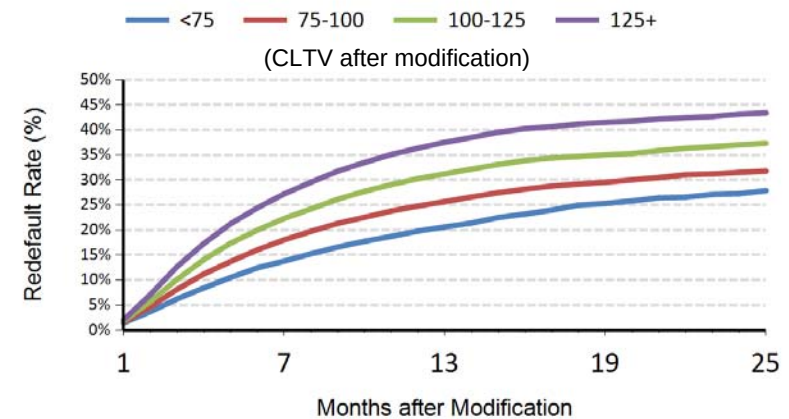
Re-Defaults for Modified/Dirty-Pay Loans

Re-Defaults by Drop in Monthly Payment



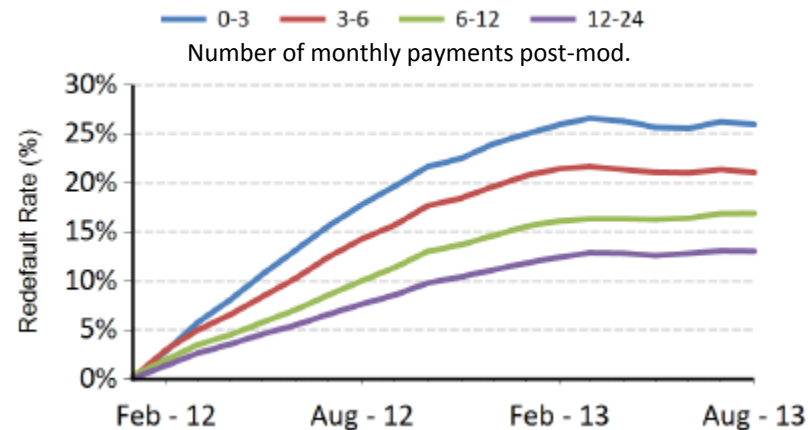
Source: Citi, CoreLogic

Re-Default Rate by CLTV



Source: Citi, CoreLogic

Re-Defaults By Payment History After Mods



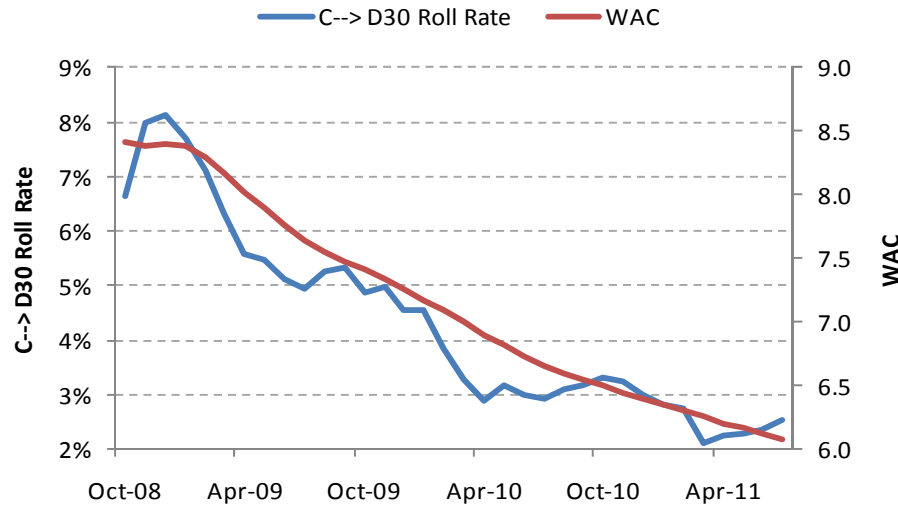
Source: Citi, CoreLogic

Re-defaults have come down significantly due to large payment cut (~ 30%) after modifications. Many principal reduction modifications are resulting in lower CLTVs and monthly payments. Modified loans are performing better than self-cures. Factors affecting re-default rate:

- Drop in monthly payment (in absence of DTI information).
- CLTV after modification.
- Payment history before modifications.
- Payment history after modification.

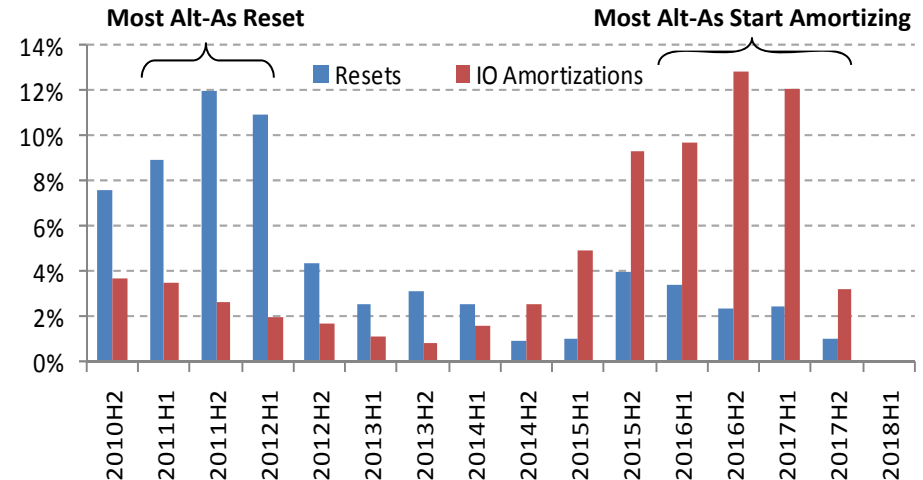
Payment Shock - Impact on ARMs and IOs

Resetting Lower ARM Coupons (Subprime ARMs)



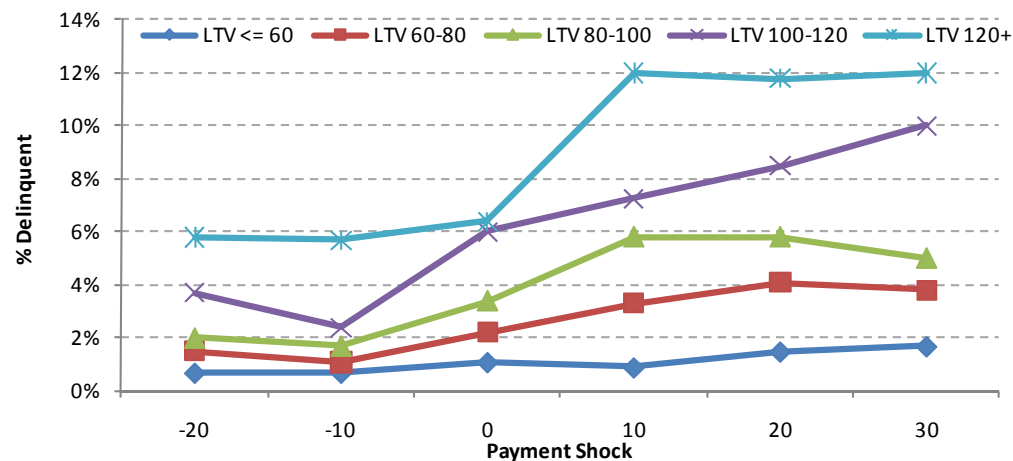
Source: Citi, CoreLogic

Resets and IO Amortizations (ALT-A ARMs)



Source: Citi, CoreLogic

Percentage of Loans That Become Delinquent Within 3 Months of Payment Shock (Alt-A ARM – 2009 Resets)

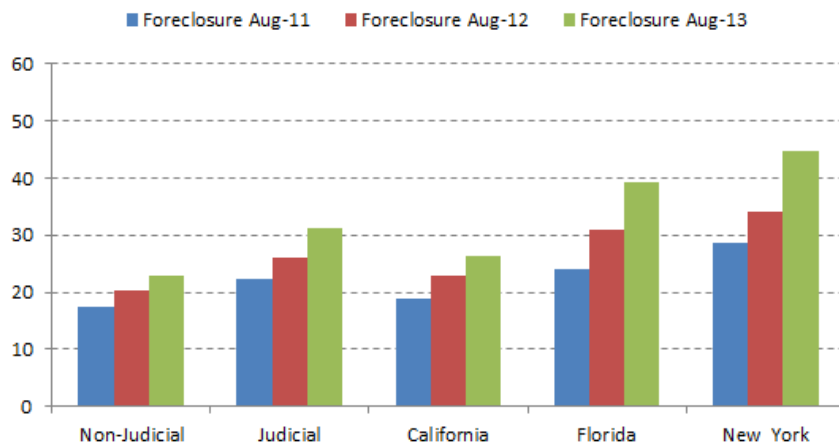


Source: Citi, CoreLogic

- Lower resetting coupons result in lower C→D30 roll rates. Alt-A hybrids have benefitted the most from lower resets as monthly payments dropped by about 50% at reset.
- However, payments will double when IO period ends posing significant default and modification risk due to payment shocks of 100% or more when they start amortizing (2015-2017). Most Alt-As had much higher monthly payment before reset.
- Little impact on prime ARMs as drop in payments due to lower resets is offset by increase in payments due to amortization (i.e., end of IO period).

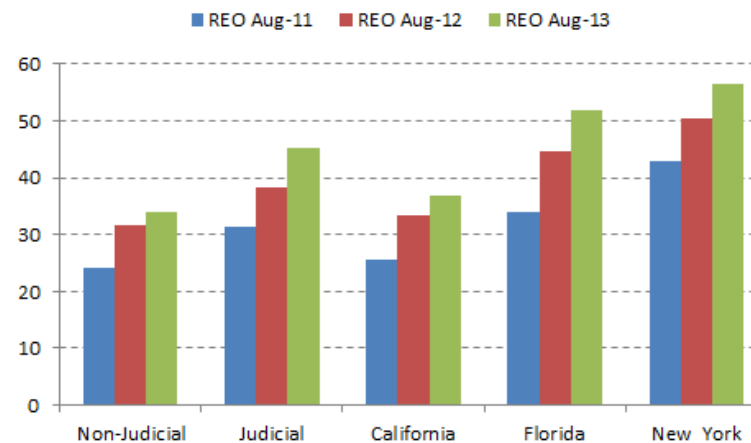
Modeling Liquidation Rates – Expect Timelines to Extend Further

Foreclosure Timelines (Months)



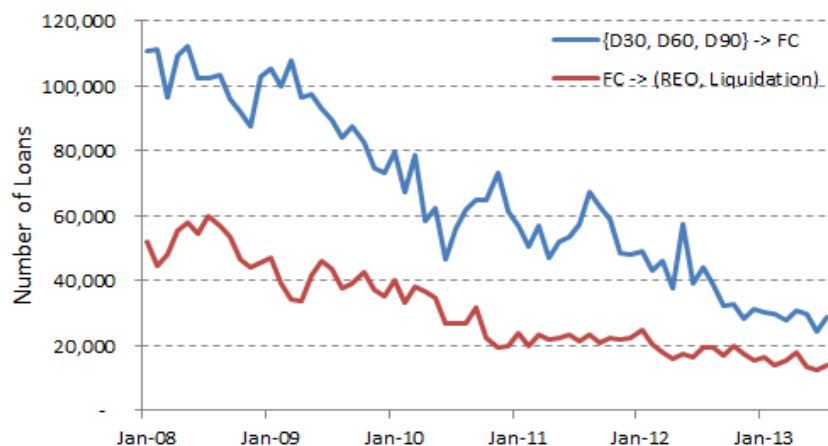
Source: Citi, CoreLogic

REO Liquidation Timelines (Months)



Source: Citi, CoreLogic

Foreclosure Backlog has Continued to Build



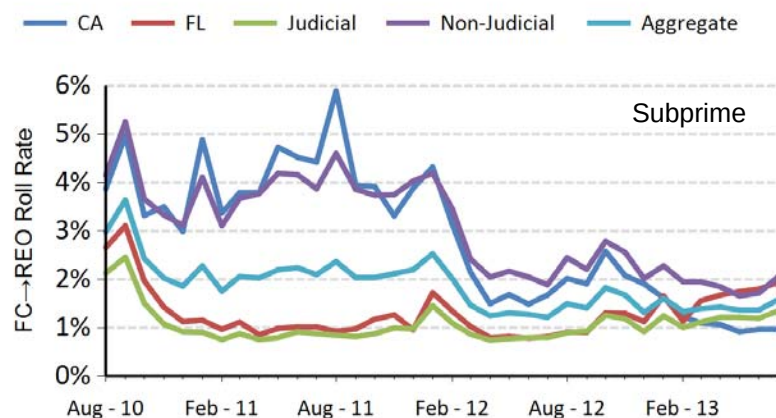
Source: Citi, CoreLogic

- More loans are added to the foreclosure pipeline each month than those that exit foreclosure through liquidation or end up in REO. This will cause the already burdened courts to be further backlogged.
- The robo-signing scandal and ensuing lawsuits has led servicers to be over-cautious. New servicing standards are stringent and have increased processing time.
- As a result liquidation timelines from foreclosure and REO continue to extend.

Slowdown in CA Foreclosure Rates Will Reduce Near Term CDRs

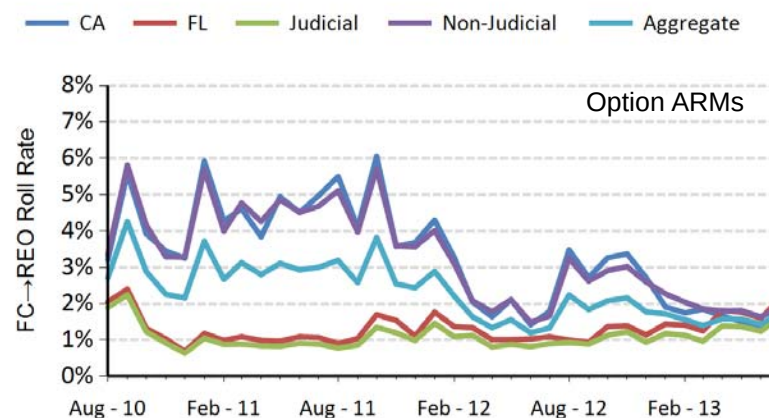


FC→REO Roll Rate by State



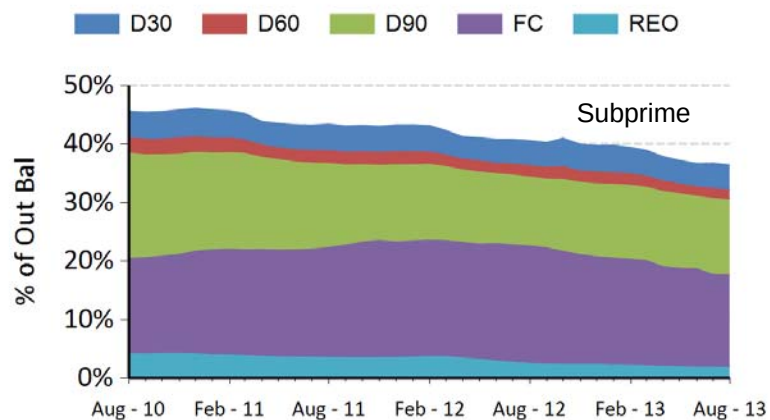
Source: Citi, CoreLogic

FC→REO Roll Rate by State



Source: Citi, CoreLogic

Distribution by Delinquency Status

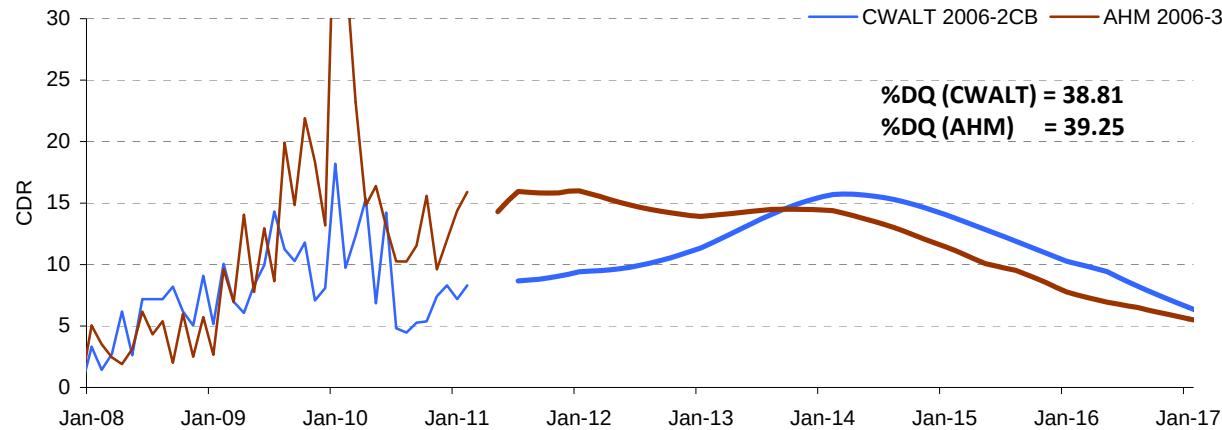


Source: Citi, CoreLogic

- Liquidation rates will continue to remain low, particularly for Option ARMs and Alt-A ARM that have a higher concentration in California where foreclosure processing rates have dropped by more than 50%.
- REO bucket that account for majority of the liquidations have continued to shrink.
- New servicing standards under the AG settlement are fairly stringent and likely to keep liquidation rates low. Most of this effect will be seen in 2014.

Deal-Specific Adjustments Capture Servicer Differences

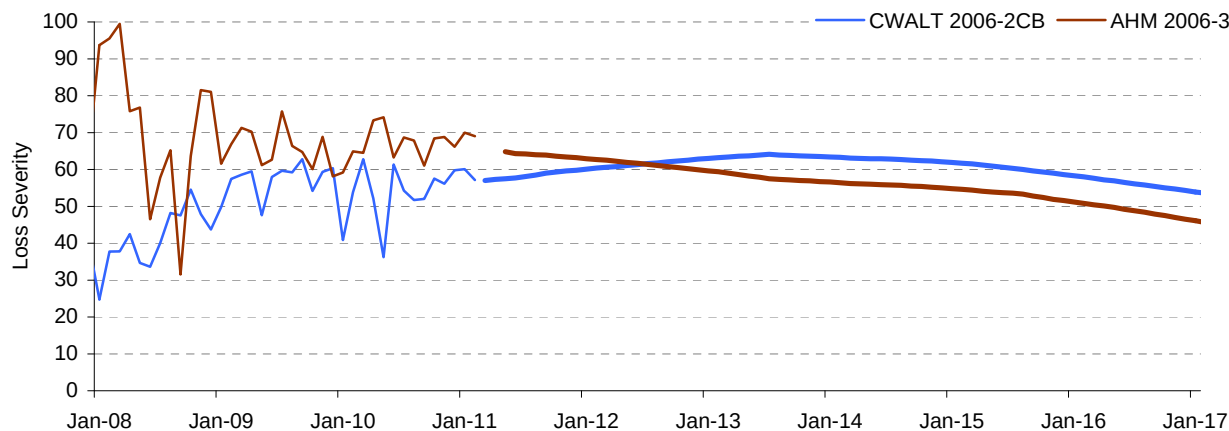
Model Projections (CDR)



Source: Citi, CoreLogic

- Despite **similar delinquencies** CDRs on CWALT is much lower than AHM. Longer foreclosure timeline should result in a more back-loaded CDR curve.
- Declining projected Loss Severity for AHM due to significant loan size differences as well as shorter foreclosure timeline.

Model Projections (Loss Severity)



Source: Citi, CoreLogic

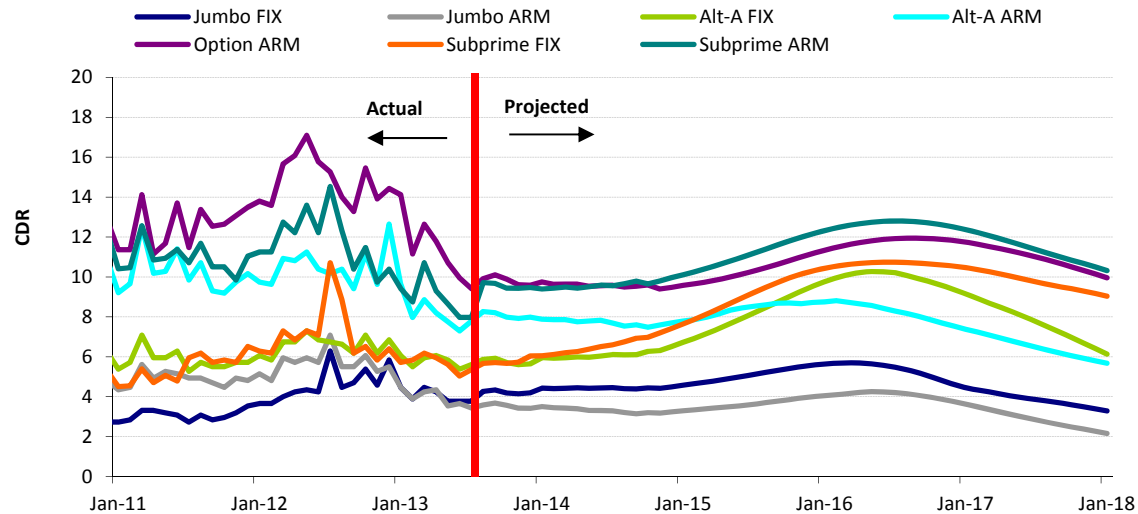
Loan Size

	DQ Loans	Def Loans
CWALT	280,022	281,495
AHM	327,489	219,361

	% Delinquent >= 24 months
CWALT	30.34%
AHM	24.03%

Expect Liquidation Rates to Peak in 2016

Model Projected CDRs by Sector



Source: Citi, CoreLogic

- Prime, Alt-A ARMs, and Option ARM CDRs are likely to continue to remain low due to a significant drop in foreclosure completion rate in California after the Home Owner Bill of Rights was passed.
- Given that over 60% of the Subprime ARM loans have been modified, we expect burnout in the new modifications in the sector. However, the pickup in liquidation rates for Subprime ARMs will not be dramatic as re-modifications will continue to exert a downward pressure. Subprime fixed-rate CDRs have been low compared to their delinquency levels and we think that their CDRs are likely to pick up sooner. They also have less concentration in California compared to subprime ARMs.
- New servicing standards under the AG settlement are fairly stringent and additional lawsuits will continue to delay foreclosure processing.

Loss Severity Example

Original House Value:	\$300,000
Loan Amount:	\$270,000
Loan Coupon:	6%
CA Home Prime Decline:	45%
Current "Market Value" of House:	$\$300,000 * 55\% = \$165,000$

Time from becoming delinquent to house sale:	16 months
Estimated REO Discount:	25%

Calculation of proceeds

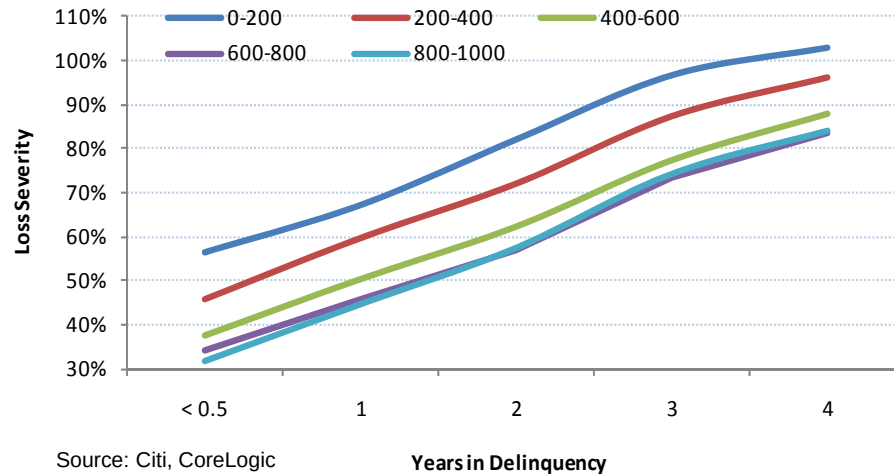
Costs and interest advances increase with increasing liquidation timelines.

Sale Price	(Market Value less REO Discount = $\$165,000 * 75\%$)	\$123,750
Interest Advances	$(\$270,000 * 6\% * 16/12)$	-\$21,600
Attorney Fees		-\$2,190
Property Taxes	$(\$5850 * 16/12)$	-\$7,800
Maintenance Costs	$(\$165,000 * 2.25\% * 16/12)$	-\$4,950
Hazard Insurance	$(\$165,000 * 0.825\% * 16/12)$	-\$1,815
Other Costs		-\$8,640
Realtor Fee	$(\$123,750 * 5\%)$	-\$6,188
Net Proceeds		\$70,568
Recovery Rate	$(\$70,568 / \$270,000)$	26.1%
Loss Severity	(100% - 26.1%)	73.9%

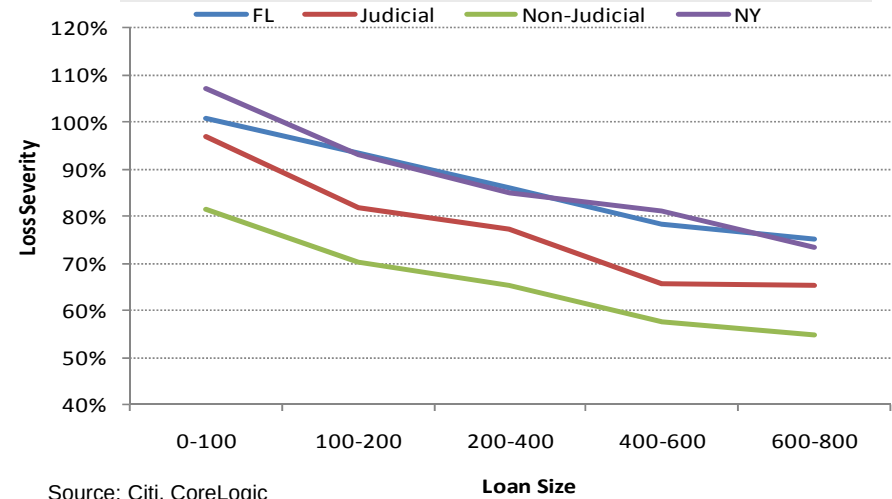
- Longer liquidations timeline result in higher discount value.

Factors Determining Loss Severity

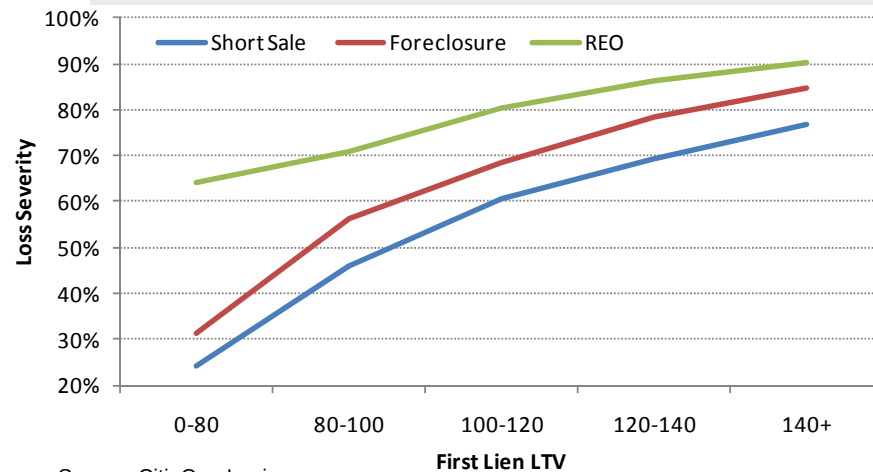
Loss Severity on REO Liquidations by Loan Size and Months Spent in Delinquency



Loss Severity on REO Liquidations by Loan Size and State



Loss Severity by Liquidation Type (200K-300K)

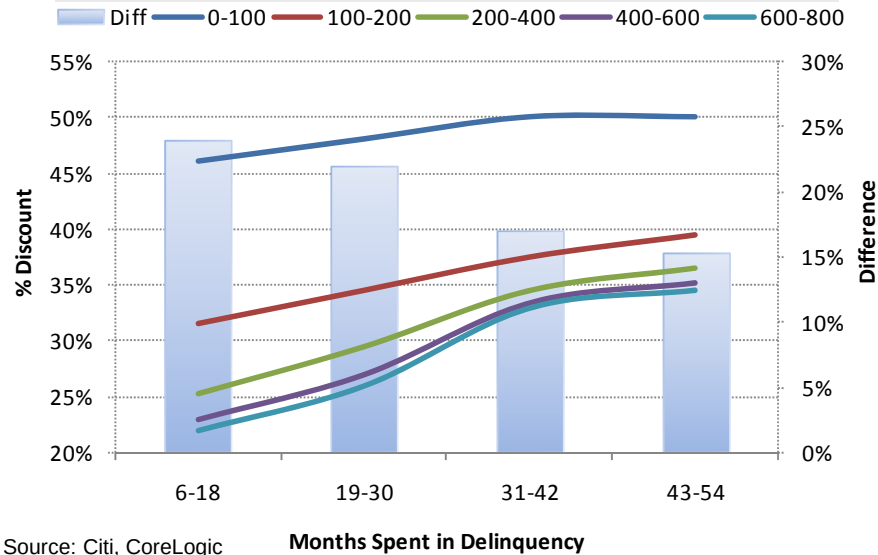


- ✓ **Loan size**
- ✓ **Current LTV**
- ✓ **Months spent in delinquency**
- ✓ **Terminal liquidation status (REO, FC, SS)**
- ✓ **Geo (Rust Belt, Judicial State, etc.)**
- ✓ **Property type**
- ✓ **Servicer Advancing**

See "A Loss Severity Model for Residential Mortgages," Citi, January 22, 2008.

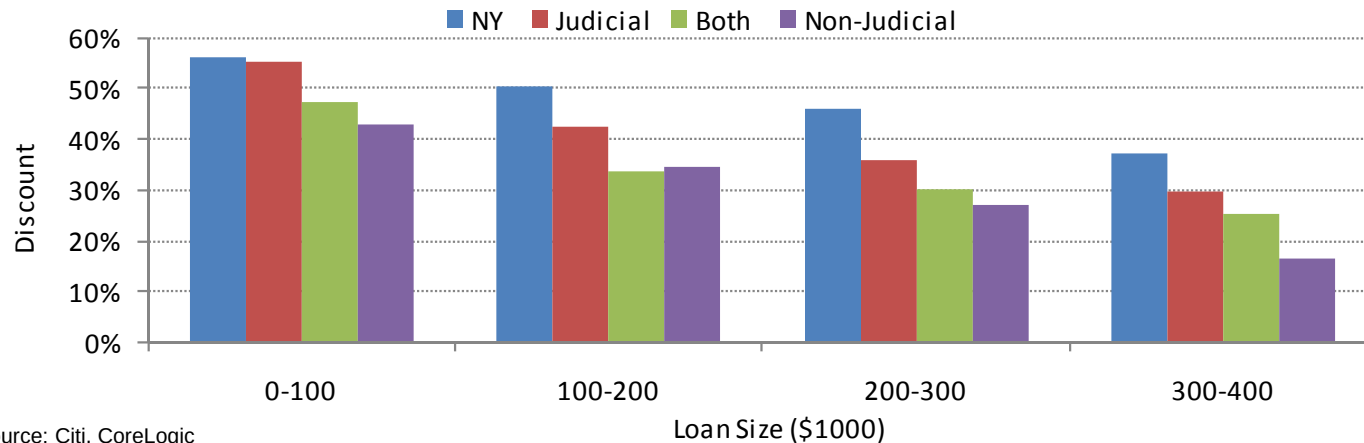
Estimating Foreclosure Discounts

Estimated Discounts by Loan Size and Months spent in Delinquency



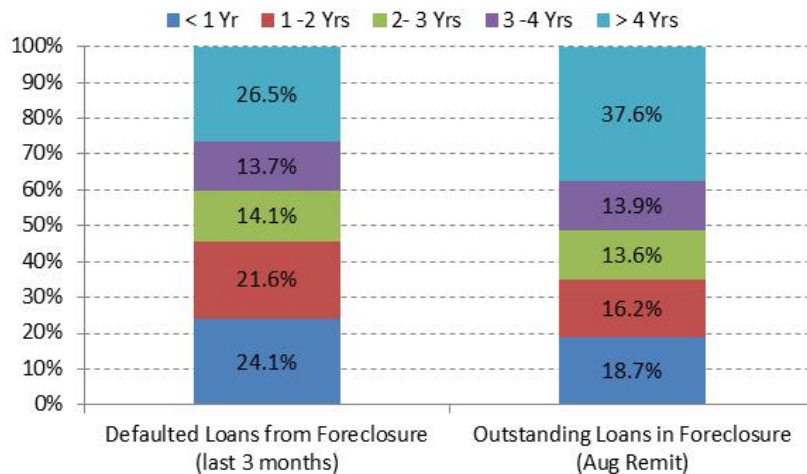
- Longer liquidation timelines result in increased costs causing severities to be high.
- In addition, property conditions deteriorate as they remain vacant and spend longer time in foreclosure.
- These discounts vary with property types (multi-family homes have higher discounts versus single-family homes) and across states (due to varying timelines and climate).
- State-level discount data is updated on a regular basis and fed to the loss severity model.

Discounts on REO sales in states with longer timelines are significantly higher



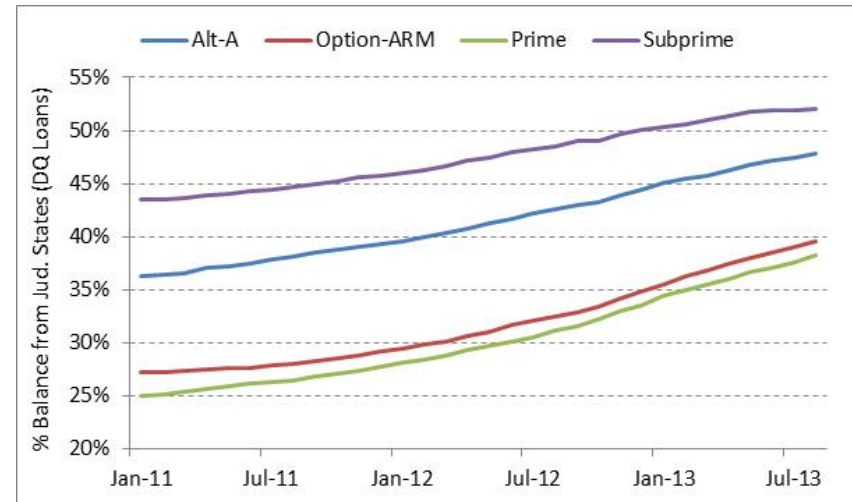
Several Factors to Put Upward Pressure on Liquidation Loss Severity

Loans in Foreclosure Have Spent More Time in Delinquency Than Those Liquidated from Foreclosure



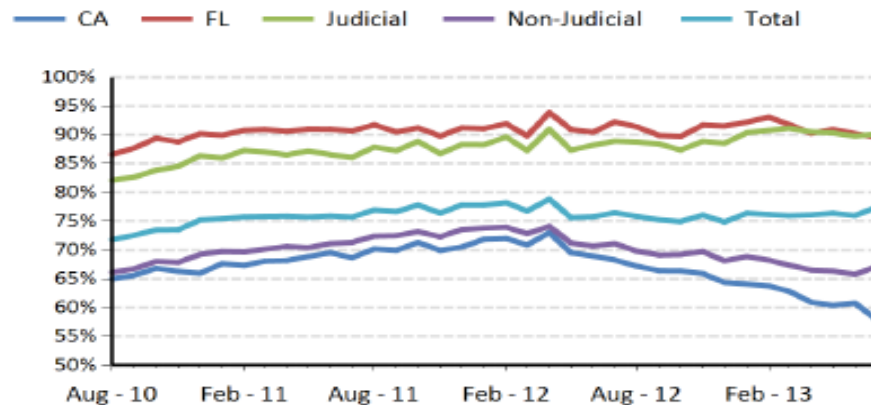
Source: Citi, CoreLogic

Share of Loans from Judicial States is Increasing in the Delinquency Pipeline



Source: Citi, CoreLogic

Loss Severity on Loans from Judicial States are Significantly Higher (Subprime)

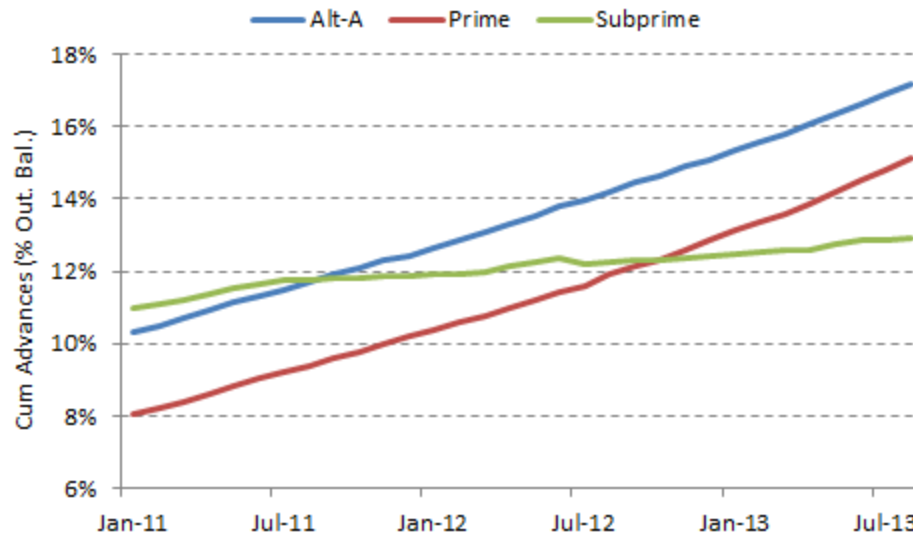


Source: Citi, CoreLogic

- We estimate that an additional one month spent in delinquency causes loss severity to increase by roughly 0.8% in terms of carrying costs.
- Delay in liquidations on loans currently in the delinquency pipeline is worse than on those that are being liquidated.
- Loans from the judicial states, that have 10-15 points higher severity, are accumulating in the delinquency pipeline. Ultimately the share of liquidations from the judicial states will increase causing loss severities to rise.

Several Factors to Put Upward Pressure on Liquidation Loss Severity

Cumulative Outstanding Interest Advances Have Continued to Increase on Delinquent Loans



Source: Citi, CoreLogic

- Due to higher advancing rate on Alt-A and Prime loans, cumulative outstanding interest advances have continued to increase. This should have a negative impact on loss severity.
- In case of subprime, advance rates have come down substantially. In addition, P&I recap mods have offset the impact of additional interest advances. Reduced advance rates should help lower loss severity.
- Despite positive home prices, loss severity has declined only marginally over the past year. In fact, liquidation severity for subprime is higher than a year ago. In New York, loss severities are up significantly despite positive HPA due to longer timelines.

Home Price Appreciation Has Not Translated into Significant Loss Severity Declines

		National		Arizona		California		Florida		New York	
		Jun-12	Jun-13	Jun-12	Jun-13	Jun-12	Jun-13	Jun-12	Jun-13	Jun-12	Jun-13
Loss Severity	Alt-A	60.2%	58.7%	53.9%	47.6%	55.2%	46.8%	77.1%	76.0%	67.4%	76.3%
	Opt-A	62.9%	58.7%	59.6%	52.4%	60.3%	52.0%	77.6%	76.4%	58.7%	73.0%
	Prime	45.6%	41.9%	44.6%	37.1%	43.6%	35.7%	59.2%	60.7%	38.3%	45.8%
	Subprime	77.4%	78.0%	69.8%	62.6%	70.4%	62.2%	91.6%	90.6%	81.6%	85.7%
% HPA (1 Yr)		12.0%		16.6%		21.5%		11.5%		3.5%	

Source: Citi, CoreLogic

Modifications and Settlements may Lower Losses

Losses Taken on Outstanding Balance as a % of Current Balance

2006 Issue

Shelf	Out Bal	% Loss Taken	% Forb Out.
FMIC	474,968,419	26.1%	1.5%
FHLT	2,572,762,607	22.5%	1.3%
CBASS	1,415,951,810	22.3%	1.4%
GSAMP	2,956,220,723	18.7%	1.3%
OWNIT	866,639,083	17.2%	0.7%
RAMC	1,213,070,603	16.2%	4.0%
MSIX	584,748,402	16.1%	0.8%
NHEL	884,116,544	15.6%	0.4%
ACE	4,325,562,307	14.7%	1.8%
SABR	2,384,721,464	14.5%	1.5%
HEAT	1,561,891,890	14.1%	0.7%
IXIS	500,201,715	13.4%	2.5%
DSLA	551,837,802	13.1%	1.0%
WMABS	558,065,816	12.3%	0.7%
MABS	2,872,687,528	12.1%	1.0%
BSMF	1,916,957,713	11.6%	6.5%
ABFC	1,222,140,723	11.4%	1.5%
RAMP	2,408,245,732	10.9%	5.1%
LBMLT	5,280,280,910	10.5%	0.5%
RASC	2,989,366,792	10.3%	3.0%
CWL	14,126,800,202	10.2%	2.5%
ABSHE	1,524,710,391	9.0%	1.3%
MSHEL	595,231,227	8.8%	0.4%
SAST	691,131,472	8.7%	0.8%
MLMI	4,386,299,728	8.5%	0.9%
BSABS	4,050,290,048	8.1%	2.2%
INABS	1,220,540,942	8.0%	0.7%
OOMLT	1,496,076,270	7.8%	2.3%
WMALT	4,429,387,199	7.7%	1.0%
SAIL	2,322,565,359	7.3%	1.1%
AHMA	2,704,099,640	7.0%	4.0%
MSAC	5,459,096,535	7.0%	0.5%

2006 Issue

Sector	Out. Bal	% Loss Taken	% Forb Out.
Alt-A	95,245,560,058	3.5%	0.9%
Option ARM	39,257,981,164	6.3%	2.0%
Prime	38,301,956,366	1.4%	0.5%
Subprime	111,599,585,154	9.3%	1.6%

Repurchase of Loans Due to R&W Issue

Filing	Pending Repurchase	
	2013Q1	2013Q3
MS	1,182,000	54,485,516
NHELI	23,380,888	23,378,144
BSABS I	3,930,488	14,907,697
DB SPI	-	13,705,802
ACE	-	13,058,664
DLJMA	2,330,550	4,378,897
JPMAC	-	2,039,900
FTN	-	901,442
GSMC	337,940	337,940
CS	255,300	255,300

Filing	Repurchased	
	2013Q1	2013Q3
MLMI	1,512,495	78,914,643
SAMI II	20,720,996	13,828,973
BSABS I	3,477,778	9,253,142
DLJMA	21,478,604	5,629,848
MS	-	3,844,646
FTN	5,890,659	1,535,740
CMLTI	1,834	446,736
WF ASC	-	379,162
CWMBS	-	67,746

- Significant losses have already been taken on outstanding loans through principal reduction, treatment of forbore principal as losses, and unpaid interest on delinquent loans.
- All these factors should help reduce losses on these loans.
- We expect loss reversal on many loans as they get repurchased due to R&W issue (or any lawsuit settlement).

Source: Citi, CoreLogic

Source: Citi, ABS-15G

This presentation has been prepared by Markets Quantitative Analysis ("MQA"), which is part of Citigroup Global Markets' sales and trading operations.

Settlements cash flow estimates

Rescap					
Shelf	GMAC Servicing Payout (\$ K)	RFC Servicing Payout (\$ K)	GMAC R+W Payout (\$ K)	RFC Servicing Payout (\$ K)	Total Payout (\$ K)
RASC	-	20,146	-	140,852	160,998
RALI	-	15,404	-	131,888	147,292
RAMP	-	14,454	-	121,951	136,405
GMACM	1,020	-	97,085	-	98,105
RAAC	-	4,273	-	24,684	28,957
RFMS2	-	289	-	19,019	19,308
RFMSI	-	2,102	-	14,883	16,986
CARR	-	840	-	7,704	8,544
LXS	596	735	45	4,745	6,120
LUM	141	194	2,731	1,805	4,872
HVMLT	1,945	1	2,021	448	4,414
DBALT	3,950	126	-	-	4,077
NAA	1,834	-	-	-	1,834
GSR	90	72	1,616	-	1,778
IMM	918	604	-	-	1,522
ARMT	161	-	1,118	233	1,511
SASC	74	24	957	-	1,055
CMLTI	574	407	-	-	981
MHL	967	0	-	-	967
CSFB	68	9	847	-	924
MASD	290	563	-	-	853
NHELI	802	-	-	-	802
GSMPs	100	-	697	-	797
Total	16,668	60,901	111,199	469,605	658,373

JPM	
Shelf	Payout (\$ MM)
BALTA	586
BSAAT	35
BSABS	900
BSARM	126
BSMF	537
BSSLT	68
BUMT	6
CFLX	60
CHASE	88
EMCM	12
GPMF	187
JPALT	205
JPMAC	531
JPMMT	157
LUM	19
MSST	30
PRIME	20
SACO	355
SAMI	578
Grand Total	4,500

Countrywide	
Shelf	Payout (\$ MM)
CWALT	3,828
CWHEL	28
CWHL	929
CWL	3,714
Grand Total	8,500

Expect More Forbearance Related Losses

2006 Issue

Shelf	Out Bal	% Forb Out.
BSMF	1,916,957,713	6.5%
CARR	3,172,074,258	5.9%
OPMAC	421,394,961	5.2%
GPMF	2,995,734,256	5.2%
RAMP	2,408,245,732	5.1%
AHMA	2,704,099,640	4.0%
RAMC	1,213,070,603	4.0%
RAAC	217,247,510	3.6%
MHL	215,332,159	3.4%
NCMT	370,900,849	3.3%
GMACM	443,872,730	3.2%
RASC	2,989,366,792	3.0%
RFMSI	2,341,488,762	3.0%
RAI	8,688,654,352	2.7%

Source: Citi, CoreLogic

- Number of deals have significant amount of forborne principal that have not been recognized as losses.
- Most of deals are those serviced by GMAC, Carrington, and Ocwen.
- While some deals have taken significant forbearance related losses in recent months, they have done new forbearance mods on which principal was not written down.

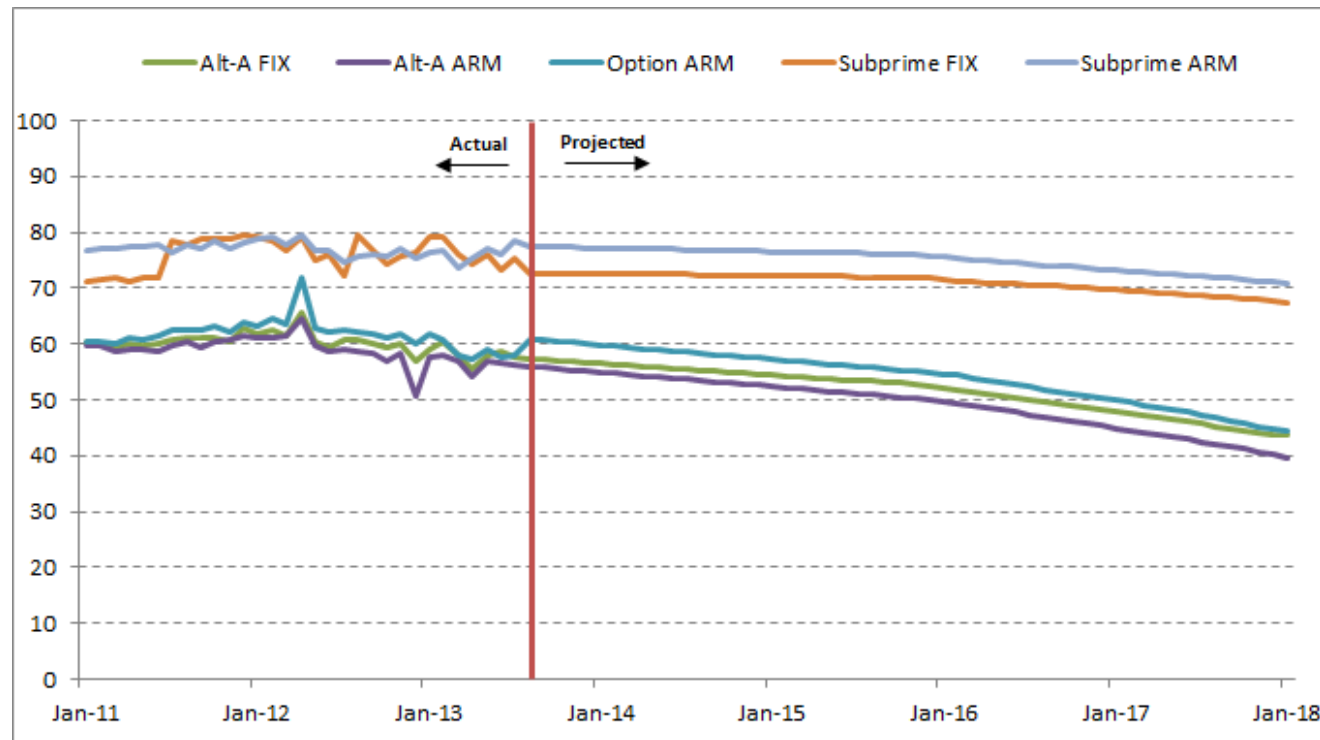
2006 Issue

Servicer	Out. Bal	% Forb Out.
GMAC MORTGAGE LLC	4,516,619,490	5.0%
CARRINGTON MORTGAGE	3,699,305,429	5.0%
OCWEN PRIOR RESCAP	13,610,585,255	3.8%
EMC MORTGAGE CORPORATION	7,863,300,316	3.1%
MIDLAND MTG	1,015,612,278	2.6%
OCWEN	30,936,278,763	2.0%

Source: Citi, CoreLogic

Liquidation Loss Severity to Remain Elevated

Model Projected Loss Severity by Sector

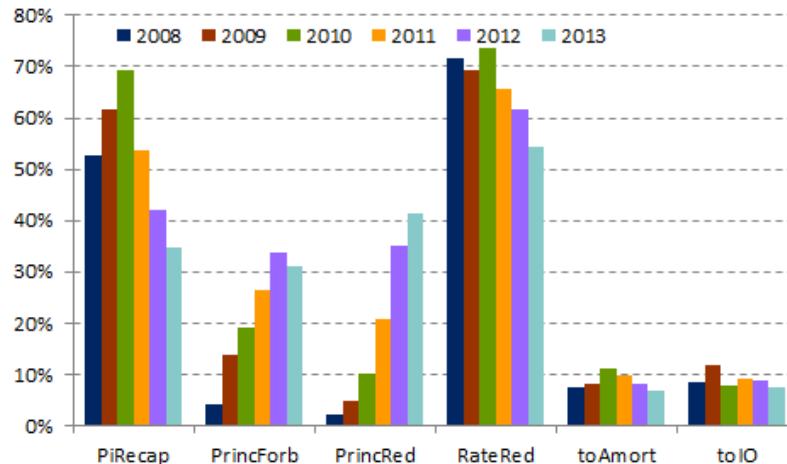


Source: Citi, CoreLogic

- We expect subprime loss severity to remain relatively elevated due to increasing concentration in judicial states, longer time lines, and adverse selection.
- Alt-A and Option ARM loss severities are likely to benefit more from home price appreciation. However, some of the declines in severity is likely to be offset by increasing outstanding advances as well as slowing liquidation rates in California. The share of loans from the judicial states has also been increasing but the percentage is lower than those for subprime.

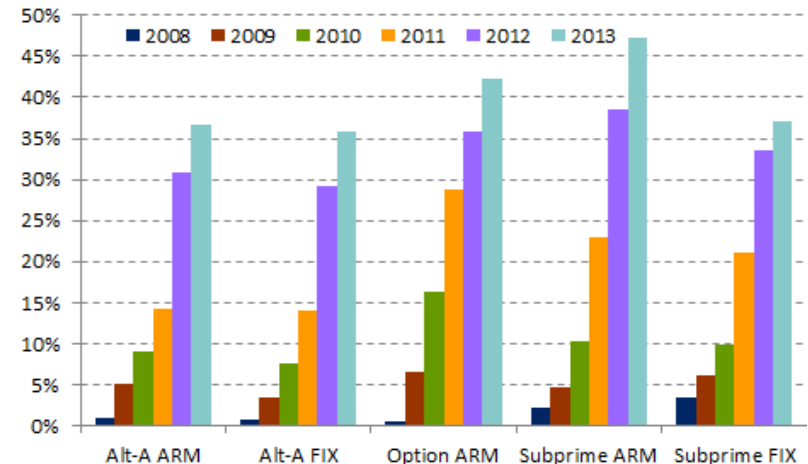
Share of Principal Reduction Modifications will Remain Elevated

Distribution by Modification Type (All Sectors)



Source: Citi, CoreLogic

Principal Reduction Modifications Have Increased

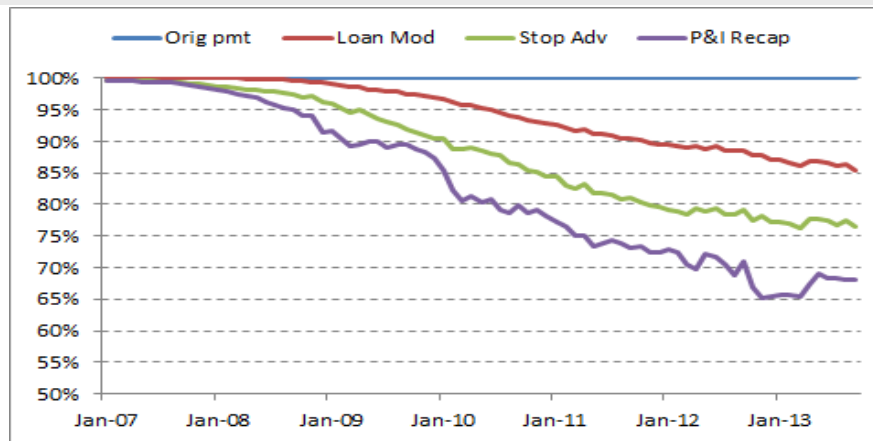


Source: Citi, CoreLogic

- Rate reductions continue to be the dominant type of loan modification comprising about 55% of all the loans modified in 2013.
- Share of principal reduction and principal forbearance modifications has been increasing, especially in the Option ARM and Subprime sectors.
- Several factors are likely to result in increased principal reduction modifications:
 - Lower coupons: With many of the ARMs resetting to lower coupons, there is less room for further decrease in the mortgage rates through loan modifications.
 - Interest-only loans: Many of the delinquent loans are IOs that are generally modified to fully amortizing loans. To keep their monthly payments low, servicers need to reduce their outstanding balance.
 - High loss severity tilts NPV in favor of loan modifications even with forgiving significant amount of principal.
 - Increasing foreclosure timelines: Servicers may be under pressure to modify these loans and often reduce principal so that the borrowers remain current and there is some cash flow.

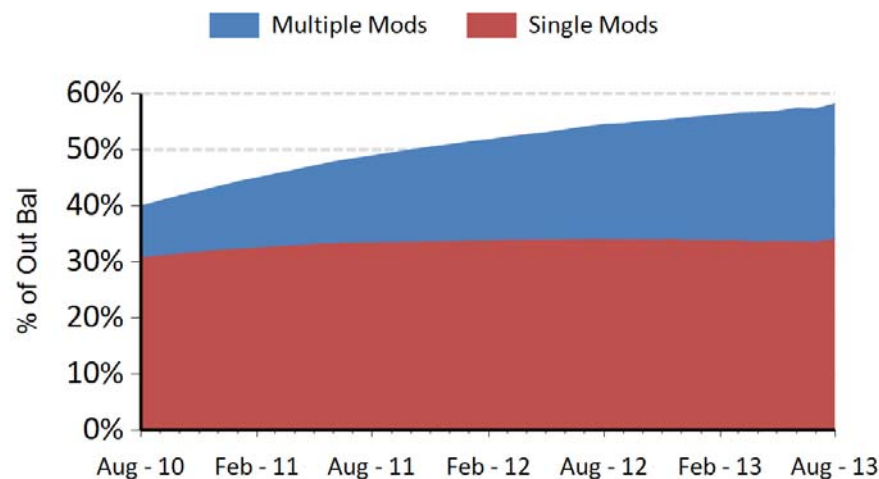
Cashflows Will Continue to Decline Due to Mods and Reduced Advancing

Comparison of Actual to Estimated Scheduled Cash Flows (Subprime)



Source: Citi, CoreLogic

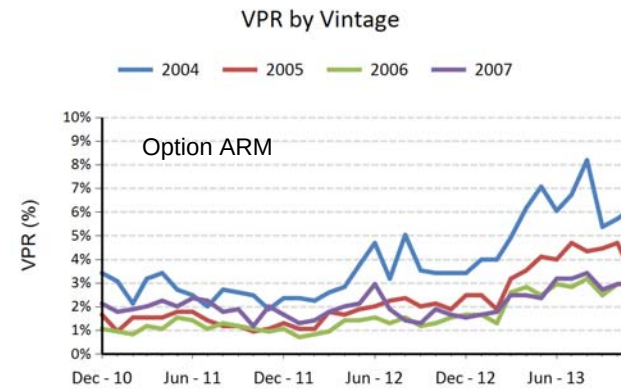
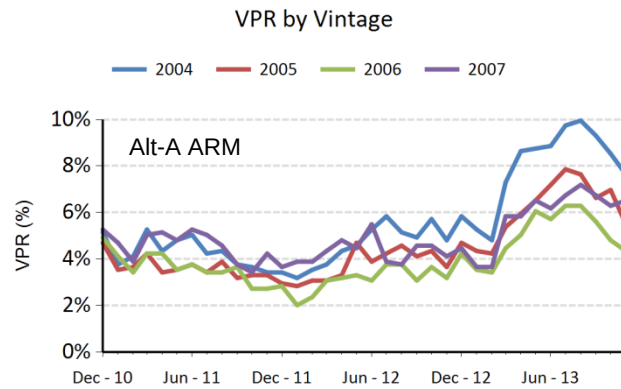
Cumulative Modifications (Subprime)



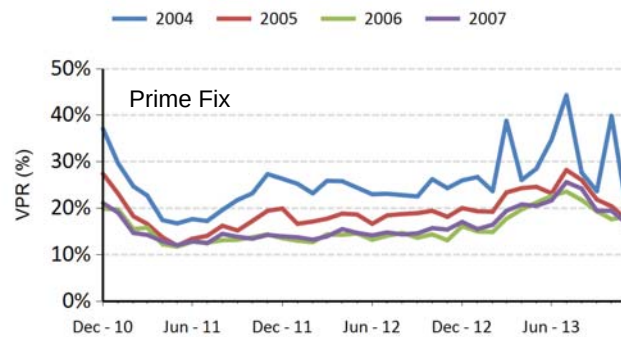
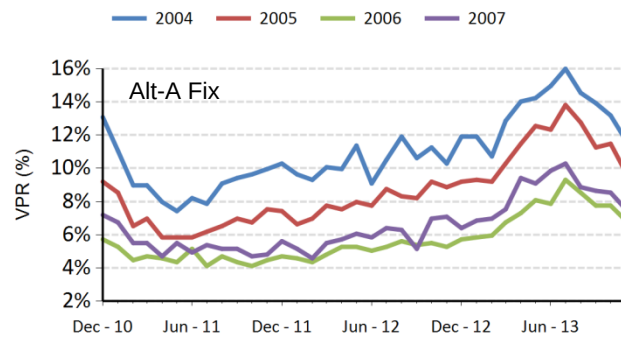
Source: Citi, LoanPerformance

- Loan modifications make cash flows uncertain that remain at variance with those implied by published prepayments and defaults.
- Detailed algorithm analyze historical monthly payment patterns for each loan and identifies: (a) different types of loan modifications, (b) change in mortgage terms, and (c) impact on principal and interest cashflows.
- A separate logic estimates P&I advancing rate.
- These data together are used to derive loan modification vectors such that projected cashflows are close to actual.
- **Overall we expect cashflows to continue to decrease due to mods (and re-mods) and reduced advancing as Ocwen and Nationstar have a larger servicing share.**

Recent Increase in Mortgage Rates will Damp Alt-A and Prime Speeds



Source: Citi, CoreLogic



Source: Citi, CoreLogic

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Model Projected VPRs

Category	VPR (1 Yr)	VPR (4 Yr)	VPR (Life)
JFX	12.1	13.6	13.5
JAR	10.0	11.7	12.0
AFX	7.6	8.0	11.5
AAR	5.6	5.8	5.9
OAR	1.6	2.4	2.8
SFX	5.9	5.9	5.1
SAR	1.3	2.0	2.3

Source: Citi

Model Projected Cum Defaults and Losses

																				Model Projections(%CB)							
																				Base Scenario				Stress Scenario			
																				2 Yr		Life		2 Yr		Life	
																				Cum Def	Cum Loss	Cum Def	Cum Loss	Cum Def	Cum Loss	Cum Def	Cum Loss
ISSUE YEAR	AMT OUT (\$BB)	FACTOR	WALA	LOAN SIZE(\$K)	FICO	%INV	ORIG LTV	CURR LTV	60++ DQ(%)	Mods (%)	IO (%)	%in IO prd	Not DQ in 2Yr(%)	LAST 3 MONTH				To Date(%OB)		Cum Def	Cum Loss	Cum Def	Cum Loss	Cum Def	Cum Loss	Cum Def	Cum Loss
														VPR	CDR	LS	Advrate										
Prime FIX																											
2004	3.9	0.11	118	292	707	10.3	70.0	52.7	8.6	6.9	1.5	1.4	85.56	24.0	0.9	33.6	94.8	1.1	0.3	4.2	0.7	7.1	1.0	6.5	1.3	10.5	2.0
2005	10.5	0.22	102	418	730	5.9	73.7	75.7	10.7	12.6	18.3	18.2	81.67	23.5	2.9	41.6	89.7	5.1	2.0	6.4	2.5	11.6	4.2	9.9	4.1	18.1	7.6
2006	14.7	0.24	91	463	731	6.0	76.8	87.6	14.0	19.5	31.0	30.8	76.78	20.4	3.8	43.0	85.3	9.0	4.0	8.7	4.2	16.8	7.7	13.0	6.4	25.5	12.9
2007	21.5	0.28	80	505	733	5.2	79.4	89.4	14.5	20.3	37.0	36.9	76.19	21.4	4.3	44.9	86.3	10.8	4.9	9.4	4.5	18.3	8.2	14.1	7.0	28.0	14.0
Total	50.6	0.24	90	449	730	6.0	76.7	83.2	13.1	17.4	28.6	28.5	78.22	21.7	3.6	42.8	87.4	8.3	3.7	8.2	3.7	15.6	6.7	12.3	5.8	23.9	11.5
Prime ARM																											
2004	12.7	0.13	114	335	730	11.1	72.6	62.0	7.6	5.5	73.2	18.9	88.04	17.6	1.6	32.4	87.5	2.6	0.8	3.6	1.1	7.0	1.7	6.5	2.2	11.5	3.6
2005	23.6	0.23	104	426	735	10.4	74.2	74.6	8.0	8.4	81.3	23.9	86.44	18.0	2.6	35.4	91.3	6.9	2.6	4.8	1.8	9.1	2.9	8.8	3.5	15.8	6.0
2006	17.4	0.27	93	490	732	10.3	75.7	86.4	10.9	14.9	90.4	39.8	81.37	14.9	3.4	41.2	88.5	14.0	6.1	7.5	3.3	13.7	5.3	13.3	6.1	23.5	10.4
2007	19.0	0.31	87	573	731	10.1	75.6	84.3	13.1	20.7	87.4	59.2	76.61	15.5	3.6	44.2	86.5	15.2	6.9	8.9	4.1	16.5	6.5	15.5	7.4	27.5	12.4
Total	72.7	0.24	99	449	732	10.4	74.6	77.7	10.0	12.7	83.6	36.1	82.94	16.5	2.9	38.6	88.7	10.0	4.3	6.3	2.7	11.8	4.2	11.2	4.9	19.9	8.3
Alt-A FIX																											
2003	5.4	0.12	126	112	715	30.9	72.7	49.7	10.3	8.6	1.7	0.0	82.81	14.9	1.5	52.2	89.6	2.2	0.9	4.1	1.4	7.1	1.9	6.9	2.5	11.4	3.9
2004	10.0	0.19	114	131	711	33.0	74.9	61.9	13.2	10.4	7.1	6.7	79.20	14.6	2.3	48.9	87.7	3.9	1.7	6.8	3.1	12.3	4.9	11.6	5.6	20.2	9.3
2005	31.2	0.30	102	183	712	19.8	76.5	80.6	16.9	15.8	29.8	29.3	73.34	12.1	3.9	52.0	86.7	10.8	5.4	9.3	5.0	19.2	9.3	16.2	8.9	31.9	17.2
2006	35.9	0.32	91	215	704	17.9	78.8	92.1	25.7	27.0	34.7	33.6	61.85	8.8	6.0	60.0	77.3	22.3	12.8	13.0	7.7	29.4	16.0	21.5	13.2	46.1	27.3
2007	27.7	0.40	81	270	706	15.1	78.0	90.4	26.1	30.2	40.7	40.4	60.90	8.4	5.5	60.2	82.6	23.8	13.4	12.1	7.0	29.6	15.6	19.7	11.8	46.8	27.2
Total	110.2	0.31	95	195	708	19.7	77.3	83.6	21.4	22.2	30.7	30.0	67.47	10.4	4.7	56.4	82.8	16.7	9.3	10.7	6.0	23.9	12.3	17.9	10.4	38.2	21.6
Alt-A ARM																											
2003	1.0	0.08	124	294	712	22.3	71.3	55.0	12.2	9.7	60.7	0.2	81.96	16.5	2.0	24.7	76.9	1.8	0.5	5.1	0.7	13.2	1.0	9.0	1.3	22.4	2.3
2004	9.5	0.14	113	236	713	23.0	79.6	74.5	12.4	13.5	75.5	40.0	80.01	9.9	3.4	39.0	61.8	6.1	2.4	7.6	2.7	17.8	4.2	13.6	5.2	28.2	8.7
2005	27.6	0.24	102	265	715	20.2	81.4	91.5	17.5	18.5	87.4	55.7	73.96	7.5	4.6	49.3	68.3	18.4	9.0	10.3	5.1	23.1	8.9	18.2	9.4	36.0	16.7
2006	31.9	0.27	91	314	710	17.4	83.1	102.3	25.3	30.4	88.7	65.5	63.05	6.5	6.0	58.7	64.1	33.8	18.7	13.7	8.0	32.5	15.3	23.3	13.9	47.7	26.0
2007	22.4	0.33	82	400	713	16.3	81.3	96.8	26.5	32.3	89.5	79.1	60.67	7.6	6.3	55.6	65.6	36.9	20.1	14.7	8.2	33.0	14.8	24.3	13.9	48.5	25.4
Total	92.4	0.26	95	302	713	18.6	81.7	94.4	21.8	25.3	86.8	62.5	67.68	7.5	5.4	52.8	65.6	26.7	14.3	12.2	6.6	28.1	12.0	20.9	11.5	42.1	21.1
Option ARM																											
2004	2.2	0.09	111	296	705	13.8	74.5	72.8	22.2	23.4	0.0	0.0	66.60	7.1	4.6	45.0	88.0	6.0	2.7	12.6	4.9	30.8	8.2	22.9	9.5	47.1	16.8
2005	21.2	0.18	101	357	700	15.9	76.3	88.9	28.6	29.7	0.0	0.0	57.46	5.2	5.7	53.2	73.8	19.2	10.3	14.2	7.0	37.3	13.7	24.6	12.7	54.6	25.0
2006	37.7	0.28	90	363	704	15.8	79.3	103.0	34.5	40.1	3.1	2.5	49.88	4.0	7.6	61.6	54.9	40.6	24.0	17.2	10.3	47.6	23.6	27.7	17.0	63.6	36.6
2007	30.0	0.38	81	381	710	14.8	78.7	100.2	34.1	40.5	30.8	29.4	49.05	3.8	8.6	60.9	55.8	44.4	26.5	18.2	10.8	47.3	24.1	28.4	17.4	64.3	37.5
Total	91.1	0.29	90	365	705	15.5	78.3	98.1	32.7	37.4	11.4	10.7	51.77	4.3	7.4	59.0	60.4	36.0	21.1	16.7	9.6	44.7	21.1	27.1	16.0	61.3	33.7
Subprime																											
2003	9.4	0.06	129	92	623	6.8	80.8	63.2	18.2	36.9	0.8	0.1	67.88	7.7	3.0	71.8	53.1	6.6	3.2	10.1	7.4	27.2	17.7	17.1	12.9	39.8	28.8
2004	26.1	0.08	115	115	628	6.7	81.2	72.3	22.1	44.1	5.9	0.3	62.35	6.8	4.0	73.4	52.1	8.4	4.7	10.6	7.8	31.7	20.2	18.2	13.7	44.1	31.4
2005	59.3	0.14	102	140	626	5.8	84.2	89.8	29.5	59.2	20.0	1.0	52.74	4.9	5.7	73.8	45.7	20.4	12.7	13.2	9.9	40.7	27.1	23.1	17.6	54.7	40.2
2006	102.7	0.23	91	148	624	5.2	85.5	101.8	34.0	65.0	16.1	2.0	47.41	4.2	6.8	79.4	46.8	38.4	27.0	14.6	11.9	46.9	35.5	24.9	20.7	62.0	50.6
2007	64.5	0.37	83	169	620	5.1	83.8	98.9	34.8	65.8	13.1	2.3	46.01	4.3	6.4	79.8	39.4	39.4	28.5	14.6	12.1	48.2	37.6	24.1	20.4	63.3	53.2
Total	262.0	0.22	95	143	624	5.5	84.2	94.0	31.4	60.8	14.7	1.6	50.50	4.8	6.1	77.4	45.5	30.4	21.1	13.7	10.9	43.6	31.9	23.4	19.0	58.1	46.2

-%OBand %CBstand for percent of original balance and percent of current balance respectively. Loss as a %OB= Loss to date + factor * (Loss as a %CB).

-The Base Scenario assumes that home prices rise 5% in the 1st year, 5% in the 2nd year, and then increase by 3% per year, while unemployment goes to 6.4% by Q4 2014.

-The Stress Scenario assumes that home prices drop 4% in the 1st year, 1

Model Projected Yields

						Cum Collat		PWD
			Balance	Px	Yield	WAL	Defaults %	
Subprime	Sequential		31,624,386,415	68.36	5.10	8.94	48.6	21.3%
	Pro-rata		86,930,012,088	60.01	5.56	14.17	47.9	32.2%
	Mezz		41,164,860,968	85.63	5.49	5.73	37.2	0.1%
POA	SSNR		69,850,569,104	70.12	6.34	7.87	46.7	17.4%
	SNR MEZZ		22,540,214,462	57.43	8.12	8.15	45.5	31.9%
Prime	2005	Total	35,225,294,898	95.75	4.95	5.08	10.8	3.6%
		FIX	11,368,125,845	97.72	4.92	4.43	12.5	4.9%
		FLT	4,866,734,680	95.58	4.67	5.47	8.6	1.4%
		WAC	18,990,434,373	94.62	5.03	5.37	10.3	3.4%
	2006	Total	33,563,269,851	90.04	5.24	6.42	16.4	12.2%
		FIX	13,793,037,536	93.16	4.97	6.44	18.1	16.4%
		FLT	4,410,468,222	85.63	5.23	7.46	15.8	11.5%
		WAC	15,359,764,092	88.50	5.48	6.10	15.0	8.7%
	Total		129,394,605,529	92.24	5.04	5.87	14.5	9.1%
	Alt-A	Total	67,160,650,155	84.68	5.34	6.25	24.1	12.3%
		FIX	25,293,869,858	89.21	5.84	5.51	21.7	14.1%
		FLT	21,254,020,499	80.07	5.01	5.88	26.3	9.2%
		WAC	20,612,759,798	83.88	5.08	7.54	24.7	13.3%
		Total	82,936,434,792	70.84	5.98	8.92	34.2	25.7%
		FIX	32,593,897,507	76.49	6.50	8.64	31.2	28.2%
		FLT	33,981,224,346	62.46	5.95	9.23	38.3	24.9%
		WAC	16,361,312,939	76.98	5.03	8.84	31.7	22.5%
	Total		236,772,862,333	77.07	5.53	7.66	30.0	19.3%

Source: Citi, CoreLogic

Valuation of New Issue Jumbo RMBS

Deal Name	Tranche	CUSIP	WAC	Balance	CPN	Non-Agency Jumbo					Agency	
						Px	Yield	WAL	OAS -Yr	CRR	PX	OAS
SEMT 13-2	A	81745MAA9	3.94	563,366,942	1.87	87.25	3.341	11.2	21	5.3	91.48	8
SEMT 13-3	A1	81745RAA8	3.81	142,839,387	2	87.44	3.446	11.3	30	5.5	91.28	10
SEMT 13-4	A1	81744YAA4	3.78	164,972,723	2.33	87.66	3.776	11.4	67	5.5	93.53	10
SEMT 13-5	A1	81745AAA5	3.81	421,935,175	2.5	89.37	3.747	11.4	60	5.6	94.75	10
SEMT 12-2	A1	81744UAA2	4.57	50,243,030	3.5	98.5	3.707	9.2	38	15.8	102.09	9
SEMT 12-4	A1	81744VAA0	4.35	68,729,832	3.5	98.2	3.743	9.5	48	10.6	102.31	4
SEMT 12-3	A1	81745FAA4	4.45	109,225,103	3.5	97.98	3.772	9.7	45	13.7	102.98	-1
SEMT 11-2	A1	81744QAA1	4.82	88,419,556	3.9	99.56	3.962	7.9	73	18	105.2	8

Source: Citi, CoreLogic

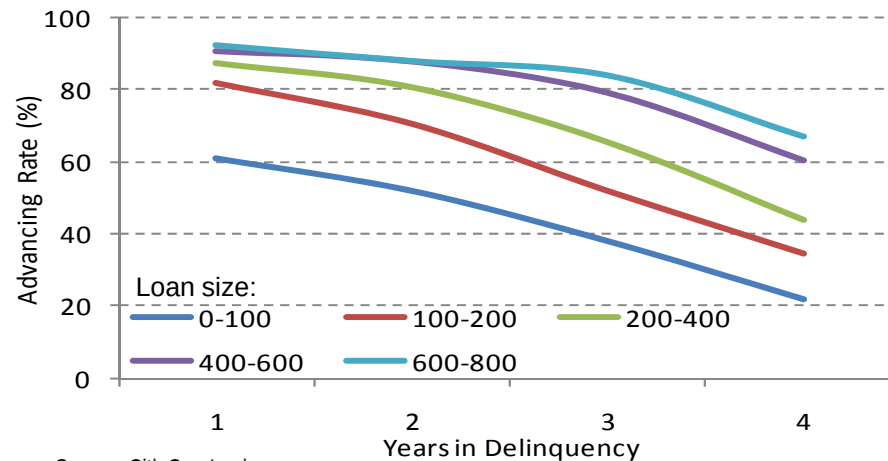
Different Types of Loan Mods and Impact on Cash Flows

Impact of Loan Mods on Cash Flows

Modification Type/Advancing	Impact on Borrower's Monthly Payment, Modified WAC, and Outstanding Principal			Impact on Deal Cash Flows		
	Monthly Payment	Modified WAC	Outstanding Principal	Principal	Interest	Net
Rate Reduction	↓	↓	-	-	↓	↓
Conversion from Amort to IO	↓	-	-	↓	↑	↓
Principal Forgiveness	↓	-	↓	↓	↓	↓
Principal Forbearance	↓	-	-	↓	↑↓	↓
Term Extension	↓	-	-	↓	↑	↓
Principal and Interest Capitalization	↑	-	↑	↓	↑	↓
Reduced Servicer Advancing	-	-	-	↓	↓	↓

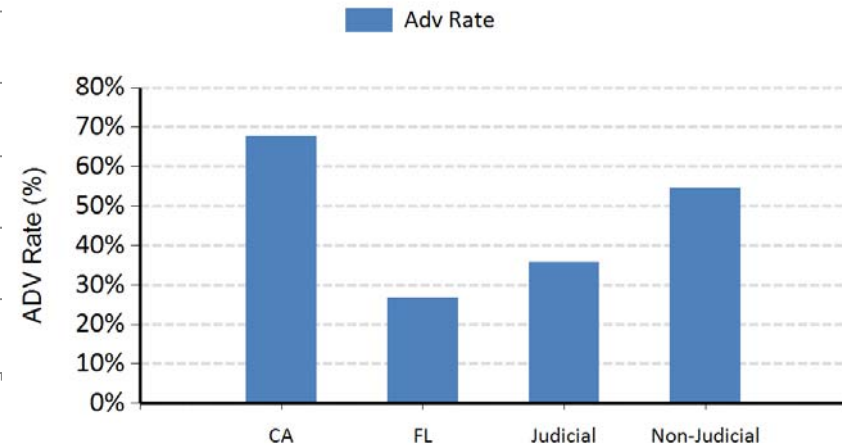
Source: Citi

Servicer P&I Advances Will Continue Declining



Source: Citi, CoreLogic

States with Long Timeline Have Lower Advance Rates (Subprime Sector)



Source: Citi, CoreLogic

- Our credit models also project vectors to adjust the projected cash flows for loan modifications, P&I recap, and servicer advances.

Estimating Haircuts and Model Adjustments

Example: HEAT 2007-2

Loan Modification Data		
	% P&I Advanced - Last 3 months	68.36%
	% Cum Modified	49.84%
MMR	Monthly Modification Rate	0.46%
PFG	% Modified With Forgiveness	57.09%
PIR	% Modified With P&I Recap Only	74.03%
RR	% Modified With Rate Reduction	66.61%
PFB	% Principal Forbearance	71.05%
TE	% Modified With Term Extension	4.40%
ΔP^{PFG}	% Principal Reduced (Forgiveness)	18.00%
ΔP^{PIR}	% Principal Increased (P&I Recap)	5.00%
ΔP^{PFB}	% Principal Forborne	24.00%
ΔR	Rate Reduced By	2.0%

Source: Citi, LoanPerformance

Historical Performance	
Vol CPRs (1 month)	3.74%
Vol CPRs (3 months)	4.44%
CDRs (1 month)	21.27%
CDRs (3 months)	15.57%
Loss Severity (1 month)	77.79%
Loss Severity (3 months)	78.84%
% Delinquent	40.91%

Source: Citi, LoanPerformance

- Loan modification, P&I advancing, and other historical data required in estimating the haircuts are available through “Col Analysis” link on Yield Book.

Principal and Interest Haircut Calculation Details

Pool Characteristics	
Outstanding Balance (\$mm)	\$449.32
Coupon (Before modification)	8.6%
WALA	51
Maturity Term	360
% of Loans Amortizing (%Amort)	87.1%

Total Est. Cash Flow (\$mm)	
Principal cash flow from voluntary prepayment	\$1.424
Principal cash flow from recovery	\$1.951
Scheduled principal	\$0.381
Total Prin Cash Flow	\$3.755
Total Scheduled Interest	\$3.106

Calculations				
Principal Haircut	Description	Formula		Value (\$mm)
(1) P&I Recap	This is the amount withdrawn by the servicer from prepayments or recoveries	Balance x MMR x PIR x ΔP^{PIR}	$\$449.32 \times 0.46\% \times 74.03\% \times 5\%$	\$0.077
(2) Advancing	This accounts for reduced principal cash flow due to loans on which the servicer is not advancing scheduled principal	Sch Prin x (%Delq) x (1 - Adv Rate)	$\$0.381 \times 40.9\% \times (1 - 68.36\%)$	\$0.049
(3) Principal Forbearance	Outstanding balance that is forborne will not amortize, thus reducing the scheduled principal	Sch Prin x (%Cum Mod) x PFB x ΔP^{PFB} x (%Amort)	$\$0.381 \times 49.84\% \times 71.05\% \times 24\% \times 87.1\%$	\$0.028
(4) Term Extension	Loans whose terms have been extended will have lower scheduled principal due. We assume the new term to be 40 years.	Difference in principal cash flow based on original and new maturity term for modified loans	This uses the amortization schedule formula	\$0.005
Principal Haircut (\$)				\$0.159
Principal Haircut (%)				4.23%

Interest Haircut	Description	Formula		Value (\$mm)
(1) Advancing	This accounts for reduced interest cash flow due to loans on which the servicer is not advancing scheduled interest	Sch Int x (%Delq) x (1 - Adv Rate)	$\$3.106 \times 40.9\% \times (1 - 68.36\%)$	\$0.402
(2) Rate Reduction	Loans whose mortgage rates have been reduced will have lower scheduled interest due	Difference in scheduled interest based on original and new mortgage rate for modified loans	This uses the amortization schedule formula	\$0.209
Interest Haircut (\$)				\$0.611
Interest Haircut (%)				19.67%

Source: Citi

Collateral Analyzer on Yield Book Provides Detailed Data

		Deal				
		C-BASS2006-CB6				
		Outstanding Loans	Delinquent Loans	Non Delinquent Loans	Defaulted Loans	Paid In Full Loans
Collateral Characteristics	Balance (MM)	\$ 251.03	\$ 118.30	\$ 132.73	\$ 230.75	\$ 271.47
	Current WAC	6.53	7.09	6.03	8.36	8.32
	Average WAC (6 months)	6.64	7.11	6.22	NA	NA
	Weighted Average Loan Age	71	70	72	38	17
	% Refinanceable	13.02%	1.46%	23.32%	0.36%	76.52%
	% Purchase Loans	22.33%	22.99%	21.74%	42.37%	19.55%
	% (CA, AZ, NV)	26.91%	25.01%	28.60%	53.30%	35.80%
	% FL	23.38%	32.22%	15.49%	21.06%	12.82%
	% (IN, MI, OH)	3.44%	3.94%	2.99%	2.39%	2.25%
	% Judicial States	44.11%	50.18%	38.69%	28.80%	33.13%
	% Full Doc	64.43%	58.43%	69.78%	48.65%	59.27%
	% Negam	0.00%	0.00%	0.00%	0.00%	0.00%
	% Investor	6.10%	6.74%	5.53%	10.91%	4.61%
	% Conforming Balance	94.64%	91.78%	97.20%	88.05%	95.57%
	Average Loan Size	172,174	192,351	157,453	200,134	175,026
	Number of Loans	1458	615	843	1153	1551
	Average FICO	618	610	625	627	601
	Service Name	LITTON	LITTON	LITTON	LITTON	LITTON
	% IO	12.69%	14.95%	10.68%	17.12%	12.37%
	% IO not amortizing	0.32%	0.38%	0.27%	15.40%	12.37%
	Remaining IO Term (months)	45.53	40.48	52.00	24.71	45.50
	% Yet to Reset	0.00%	0.00%	0.00%	22.44%	63.58%
	Average Time to Reset (month)	NA	NA	NA	7.32	16.38
	Cumulative HPA (Since Origination)	-31.20%	-34.33%	-28.41%	-30.34%	-2.75%
	Original CLTV	78.37	77.65	79.01	82.52	74.37
	Original First Lien LTV	78.18	77.52	78.76	81.36	73.67
	Average Curr CLTV (As of Last Reported Date)	113.15	120.72	106.39	121.69	76.47
	Average First Lien Curr LTV	112.85	120.55	105.99	120.14	75.73
	% Curr_CLTV < 80 (As of Last Reported Date)	15.73%	11.11%	19.86%	4.21%	52.34%
	% Curr_CLTV < 90 (As of Last Reported Date)	28.31%	21.27%	34.58%	11.92%	78.70%
	% Curr_CLTV > 100 (As of Last Reported Date)	60.12%	68.51%	52.65%	77.22%	7.39%
	% Curr_CLTV > 120 (As of Last Reported Date)	38.35%	46.71%	30.91%	50.83%	1.62%
Modification Data	% 2nd Lien	0.90%	0.59%	1.17%	5.54%	3.42%
	% With Reported Attached 2nds	0.00%	0.00%	0.00%	0.00%	0.00%
	% With Estimated Silent Attached 2nds	7.77%	9.67%	6.08%	16.14%	6.16%
	% Delinquent	47.12%	100.00%	0.00%	99.36%	12.86%
	% Delinquent >= 12 months	28.84%	61.21%	0.00%	60.71%	0.18%
	% Delinquent >= 24 months	21.74%	46.13%	0.00%	14.68%	0.07%
	% Current (Delinquent in Past 2 Years)	22.55%	0.00%	42.65%	0.00%	0.67%
	% Not Delinquent in 1 Year	42.95%	0.00%	81.23%	0.64%	86.72%
	% Not Delinquent in 2 Years	30.32%	0.00%	57.35%	0.64%	86.48%
	% Not Delinquent in 3 Years	27.63%	0.00%	52.25%	0.64%	86.46%
	% Never Modified and Not Delinquent in 2 Years	23.03%	0.00%	43.51%	0.00%	75.29%
	% Never Delinquent and Never Modified	22.26%	0.00%	42.06%	0.00%	75.29%
	% P&I Advanced - Last 3 months	32.88%	32.88%	NA	NA	NA
	% P&I Advanced - Last 6 months	33.84%	33.84%	NA	NA	NA
	% Cum Modified	48.69%	48.95%	48.46%	34.78%	0.00%
	% Cum Modified - Last 3 months	1.90%	1.78%	0.13%	NA	NA
	12 month redefault rate	27.34%	27.34%	27.34%	27.34%	27.34%
	24 month redefault rate	30.81%	30.81%	30.81%	30.81%	30.81%
	% Modified With Forgiveness	19.87%	20.37%	19.43%	10.79%	0.00%
	% Modified With P&I Recap	30.33%	32.56%	28.34%	11.62%	0.00%
	% Modified With Rate Reduction	42.15%	41.73%	42.53%	31.72%	0.00%
Dist. By Type	% Principal Forbearance	21.12%	17.26%	24.55%	9.04%	0.00%
	% Modified With Term Extension	3.06%	1.76%	4.21%	0.00%	0.00%
	% Principal Reduced (Forgiveness)	10.15%	5.61%	14.39%	10.13%	0.00%
	% Principal Increased (P&I Recap)	4.14%	4.98%	3.29%	3.22%	0.00%
	% Principal Forborne	19.87%	20.37%	19.43%	10.79%	0.00%
	Rate Reduced By	2.81	2.21	3.34	2.67	0.00
	% P&I Advanced - Last 3 months	32.88%	32.88%	NA	NA	NA

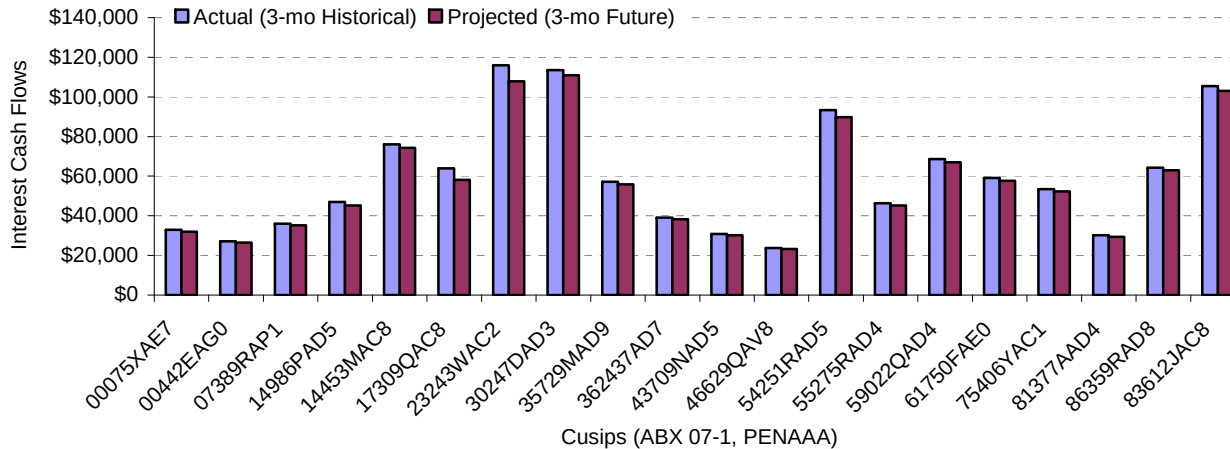
- Collateral Analyzer on Yield Book shows detailed breakdown of the collateral data both at the deal and group levels.
- In addition to the collateral data, modification and advancing data provide better understanding of actual cashflows.
- Performance data for each group is available and includes important fields such as clean-pay roll rates, re-default rates, etc.
- Includes functionality to compare simultaneous deals/cusips side-by-side.

Source: Citi, CoreLogic

This presentation has been prepared by Markets Quantitative Analysis ("MQA"), which is part of Citigroup Global Markets' sales and trading operations.

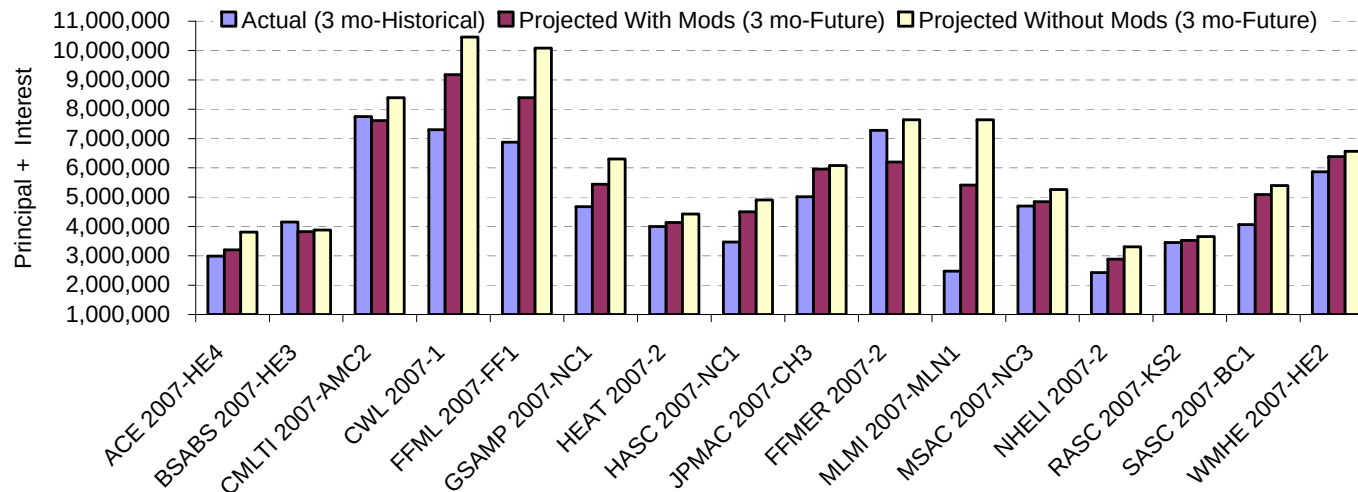
Projected Cash Flows are Close to Reported

ABX 07-1, PEN AAA Bond Level Cash Flow



Source: Citi

ABX 07-1, Deal Level Cash Flow



Source: Citi

- Principal and Interest cash flows are adjusted based on historical loan modification trends for the deal/group.
- We have our own algorithms/logic to infer different types of loan mods and the change in mortgage terms based on the monthly collateral data.

Stochastic Home Price Models and Credit-Adjusted OASs



- For evaluating prepayment risk, we combine term structure models with prepayment models => OAS analysis;
- Term structure models are **not** predictive models of interest rates. They are stochastic models that generate interest rate paths which resemble historical interest rate movements and which are consistent with market prices;
- We present here a corresponding stochastic model for HPA rates. It is not a predictive model, but does generate HPA paths that are consistent with historical behavior;
- Main difficulty is the lack of purely HPA-driven market instruments with which to calibrate the model;
- The stochastic HPA model is combined with a 2-factor term structure model to get cash flows adjusted for both prepay risk and credit risk => credit-adjusted OAS.

A Stochastic HPA Model

Key Assumptions:

- Changes in home prices are correlated with those in previous months;
- Changes in mortgage rates influence home prices, but the effect dissipates over time;

Model: Home price appreciation in month t , $h(t)$, is:

$$h(t) - \mu(t) = \alpha [h(t-1) - \mu(t-1)] + \rho [MRATIO(t) - 1] + \sigma Z(t)$$

where

$\mu(t)$: The projected equilibrium HPA for month t ;

$MRATIO(t)$: Average of past mortgage rates/current mortgage rate;

σ : Volatility of home prices

$Z(t)$: Random shock

Calibration: The $\mu(t)$ can be calibrated to the C-S or Radar Logic (RPX) home price futures. The parameters α , ρ and σ are estimated using historical data.

RFMSI 2006-S8 - Basic Information

Ch 2 Pg 4		Individual Securities Indicative Data – Non-agency CMO						
Search	Structure	P / Y	ROR	CashFlows	Hedge	WAL Sens	Act / Proj	
Recall	Clear	Performance	Col Summary	Geographic	Col History	GeoPrepay		
History		Hist Data	Comments	Col Analysis	Modify Coll	LossMatrix		
Abbrev		Manager		Issue Date		Original Amt		Trading Quality
RFM06.S8		BNP		09/28/2006		416,270,542		
5 Pieces	Group PT	05/01/11		☒ Curr		Up	Down	
Type	Cpn	Wac	WAM	Age	LIV	FICO	Pct	
1 W30	6.256	6.581	25-3	4-9	99.2	731	66.5	
2 W30-10IO	6.298	6.624	25-3	4-9	107.8	731	27.4	
3 W30-10IO-36P	6.343	6.673	25-4	4-8	123.4	748	2.0	
Delinq Status		Triggers		States		⊙ Group ○ Deal		
30 days	1.8	Delinquency Trigger		CA	38.4	CPR	CDR	IS
60 days	1.0	CumLoss Trigger		VA	8.8	1 mo	18.8	5.3 40.06
90 days +	2.7	Stepdown Trigger		FL	8.4	3 mo	17.0	7.8 49.89
In FC	7.1	60 days+ delinquencies		NY	5.7	12 mo	18.9	4.4 39.67
Bankruptcy	0.0	Cum Loss (M)		NJ	5.2	Life	15.2	
REO	0.3	13,476		Deal Cum Loss (M)		13,476		
Bonds	No 27	Paydown 05/25/11		Up	Down	<	>	
Ratings	Current Cr Supp	Orig Cr Supp	Coupon	Orig Face	Curr Face	Cumm Losses		
A1	CCC/Caa1	1.3	3.8	5.500	34,400	13,517	0.0	
A2	CCC/Caa1	1.3	3.8	0.494	25,000	6,837	0.0	
A3	NR/Caa1	1.3	3.8	7.006	(25,000)	(6,837)	0.0	
A4	NR/Caa1	1.3	3.8	6.000	(2,866)	(1,126)	0.0	
A5	CCC/C	1.3	3.8	6.000	2,600	2,600	0.0	
A6	B-/Caa1	1.3	3.8	6.000	37,400	37,400	0.0	
A7	CCC/Caa1	1.3	3.8	0.000	6,250	1,709	0.0	

Source: Citi

Analysis of RFMSI 2006-S8 A6: $HPA = (-4\%, -1\%, 0, 3\%+)$

Individual Securities											Ch 2																				
CMO Bond P/Y											Pg 4																				
Settlement Date		06/01/2011		Calculate		MatrixPY		ScenYld		FFIEC		Setup		Save		EMail															
Curve		☐ Tsy ☒ Swap		Modify		RECAL		Live		☐ AutoLive		Vol		2skw																	
Years	0.25	0.50	1.0	2.0	3.0	4.0	5.0	7.0	10.0	20.0	30.0																				
Swps	0.254	0.294	0.378	0.710	1.160	1.602	2.003	2.634	3.219	3.895	4.036																				
SwSpd	0.254	0.294	20.7	16.7	27.0	25.3	19.4	18.9	9.0	18.9	-24.6																				
Pricing Level				Macro Scenario				Prepayments				OAS Model																			
Price				95				HPA				Unemploy				Model				100				☐ OAS				☐ Eff Dur			
Defaults				Loss Severity				Delinq/Stepdown																							
Model				100				Model				100				Percent															
Lag				Haircut: Prin				0				Int				0				☒ Prin & Int Adv											
Flat Price				95.00000				Option Model																							
Accrued				0.00000				OAS																							
Full Price				95.00000				Convexity Effect																							
Nominal Measures								Credit Loss (Coll Loss)																							
Yield				5.422				Credit Loss (PWD/ISF)																							
Duration				4.987				Eff Duration																							
Discount Margin								Eff Convexity																							
Spread				iCurve				Forward Margin																							
WAL				Balance Drop				Forward WAL																							
Prin/PWD Window				10/11-09/36				Model Projections				Details																			
Prin Writedown				% Curr				9.80				1 Year				Long Term															
Shortfall				Total				0				VPR				13.0				9.8											
Min Credit Support				0.00								CDR				4.5				4.6											
Call Measures												Loss Severity				42.0				52.5											
Yield				5.320												% Orig				% Curr											
WAL				5.179												Cum Defaults				26.8											
Calculated Call				05/25/2020												Cum Loss				9.5				14.1							

Source: Citi

This presentation has been prepared by Markets Quantitative Analysis ("MQA"), which is part of Citigroup Global Markets' sales and trading operations.

Analysis of RFMSI 2006-S8 A6 for Three HPA Scenarios

HPA Assumption	(0,0,0, 3%+)	(-4%, -1%, 0, 3%+)	(-10%, -10%, 0, 3%+)
Cum Defaults (% Curr Bal)	24.9	26.8	31.7
Cum Losses (% Curr Bal)	12.6	14.1	18.5
Avg Loss Severity (%)	50.6	52.6	58.1
Bond PWD (% Curr Bond Bal)	8.3	9.8	14.7
Yield	5.65%	5.42%	4.79%
WAL (years)	6.1	6.5	7.7
Spread (bp)	349	311	217

Source: Citi

Limitations of Static Analysis: Does not evaluate the MBS over the full spectrum of interest rate and home price paths that can occur over the life of the security.

C-A OAS Analysis of RFMSI 2006-S8 A6

Individual Securities											Ch 2						
CMO Bond P/Y											Pg 4						
Settlement Date		Calculate		MatrixPV		ScenYld		FFIEC		Setup		Save		Email			
06/01/2011																	
Curve		☐ Tsy		☒ Swap		Modify		Reset		Live		☐ AutoLive		Vol		2skw	
Years	0.25	0.50	1.0	2.0	3.0	4.0	5.0	7.0	10.0	20.0	30.0						
Swps	0.254	0.294	0.378	0.710	1.160	1.602	2.003	2.634	3.219	3.895	4.036						
SwSpd	0.254	0.294	20.7	16.7	27.0	25.3	19.4	18.9	9.0	18.9	-24.6						
Pricing Level				Macro Scenario				Prepayments				Distribution					
Price				95				HPA				Unemploy					
								Model				100					
												☒ OAS					
												☒ Eff Dur					
Defaults				Loss Severity				Delinq/Stepdown									
Model				100				Model				100					
Lag				Haircut: Prin				0				Int					
												0					
												☒ Prin & Int Adv					
Flat Price				95.00000				Option Model									
Accrued				0.00000				OAS				206					
Full Price				95.00000				Convexity Effect				-27					
Nominal Measures								Credit Loss (Coll Loss)				150					
Yield				5.422				Credit Loss (PWD/ISF)				165					
Duration				4.987				Eff Duration				7.575					
Discount Margin								Eff Convexity				0.235					
Spread				iCurve				Forward Margin				233					
WAL				Balance Drop				Forward WAL				9.77					
				6.53													
Prin/PWD Window				10/11-09/36				Model Projections				Details					
Prin Writedown				% Curr				VPR				1 Year					
				9.80								Long Term					
Shortfall				Total				CDR				13.0					
				0								9.8					
Min Credit Support				0.00				Loss Severity				4.5					
												4.6					
								Cum Defaults				42.0					
												52.5					
Call Measures								Cum Loss				% Orig					
Yield				5.320								% Curr					
WAL				5.179								26.8					
Calculated Call				05/25/2020								14.1					

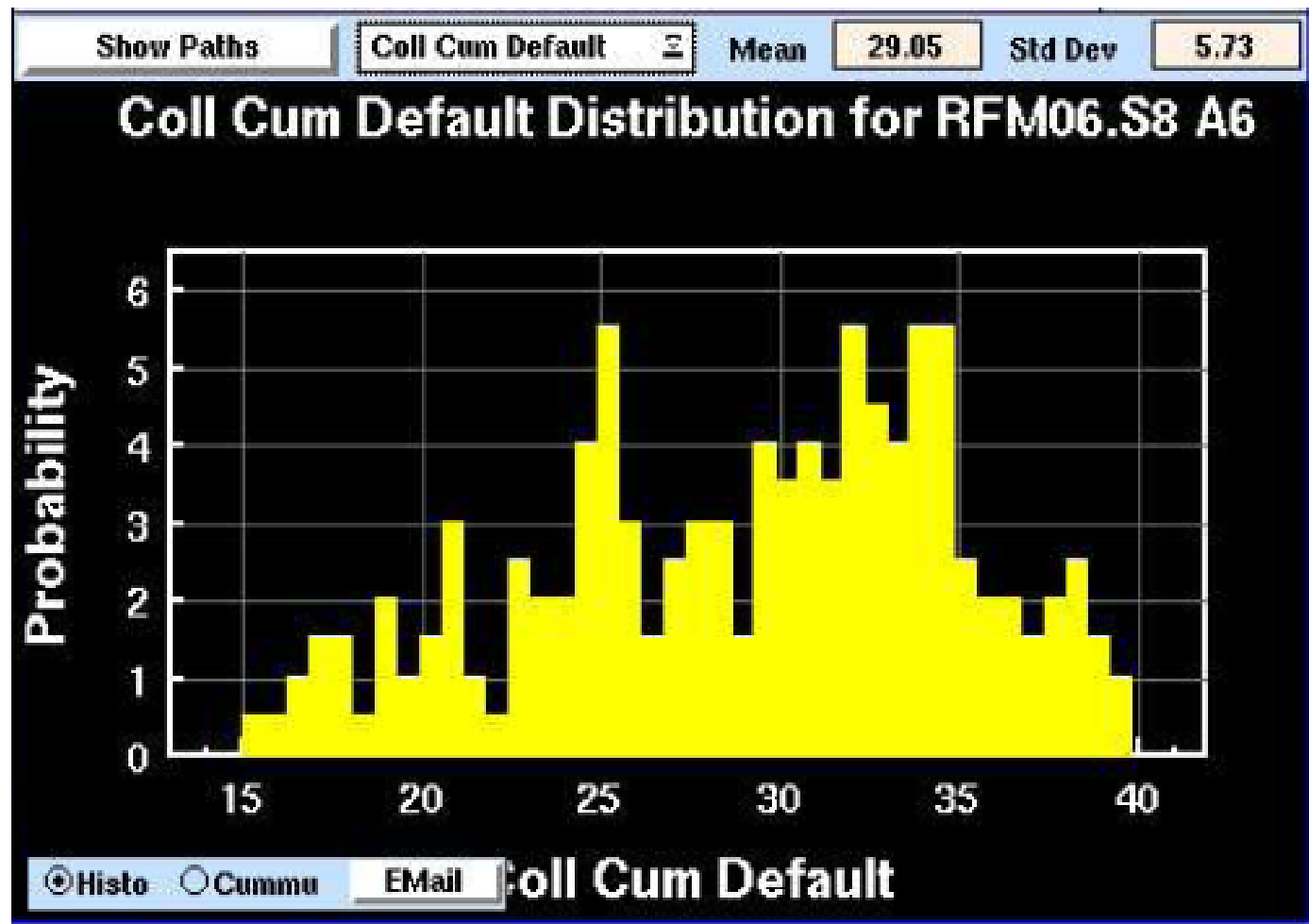
Measuring Credit Cost

Coll Loss: Change in OAS if LS=0

PWD/ISF: Change in OAS if PWDs and interest short-falls added back to bond CFs

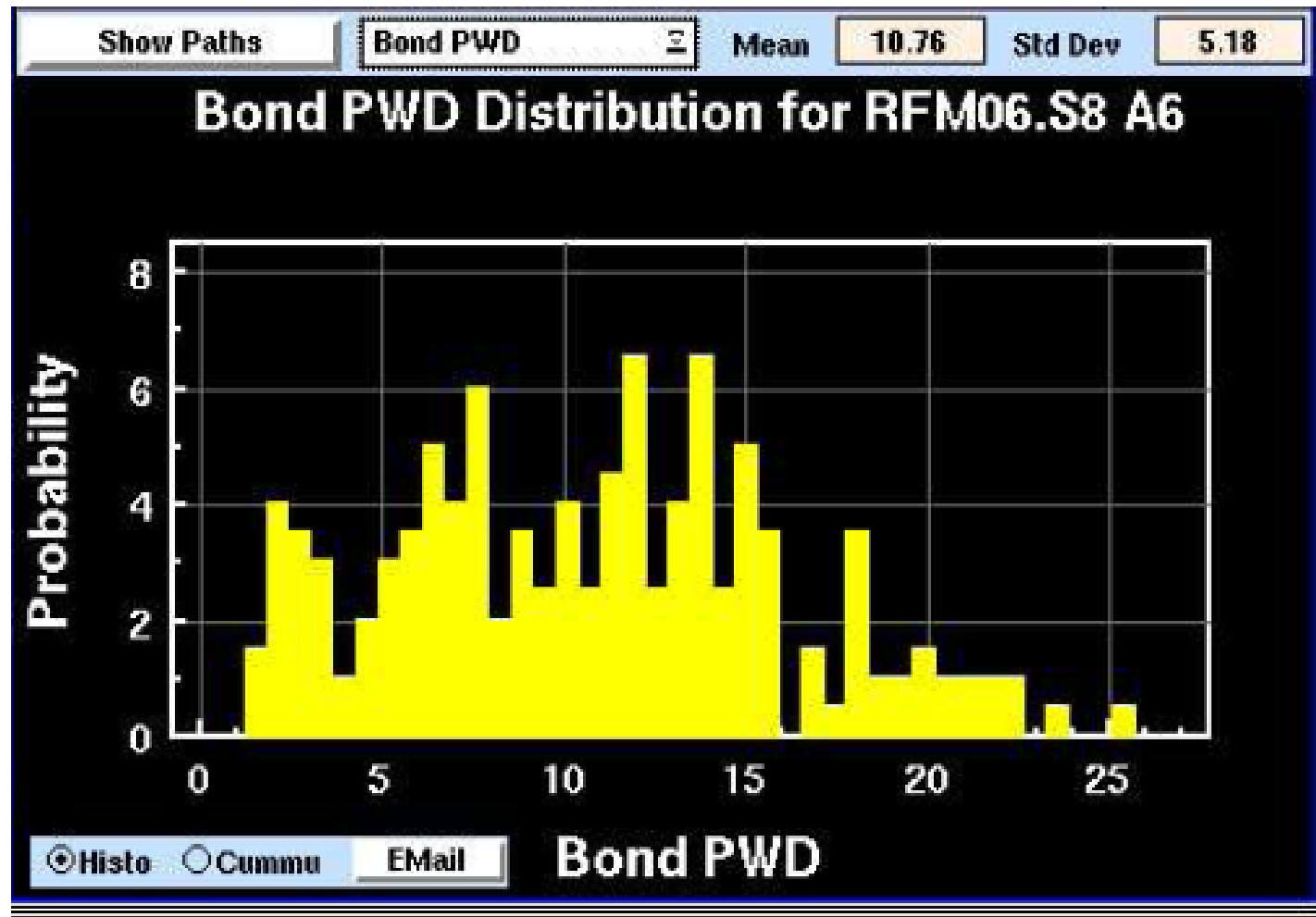
Source: Citi

Distribution of Collateral Cumulative Defaults (%CB)



Source: Citi

Distribution of PWDs for RFM06.S8 A6 (%CB)

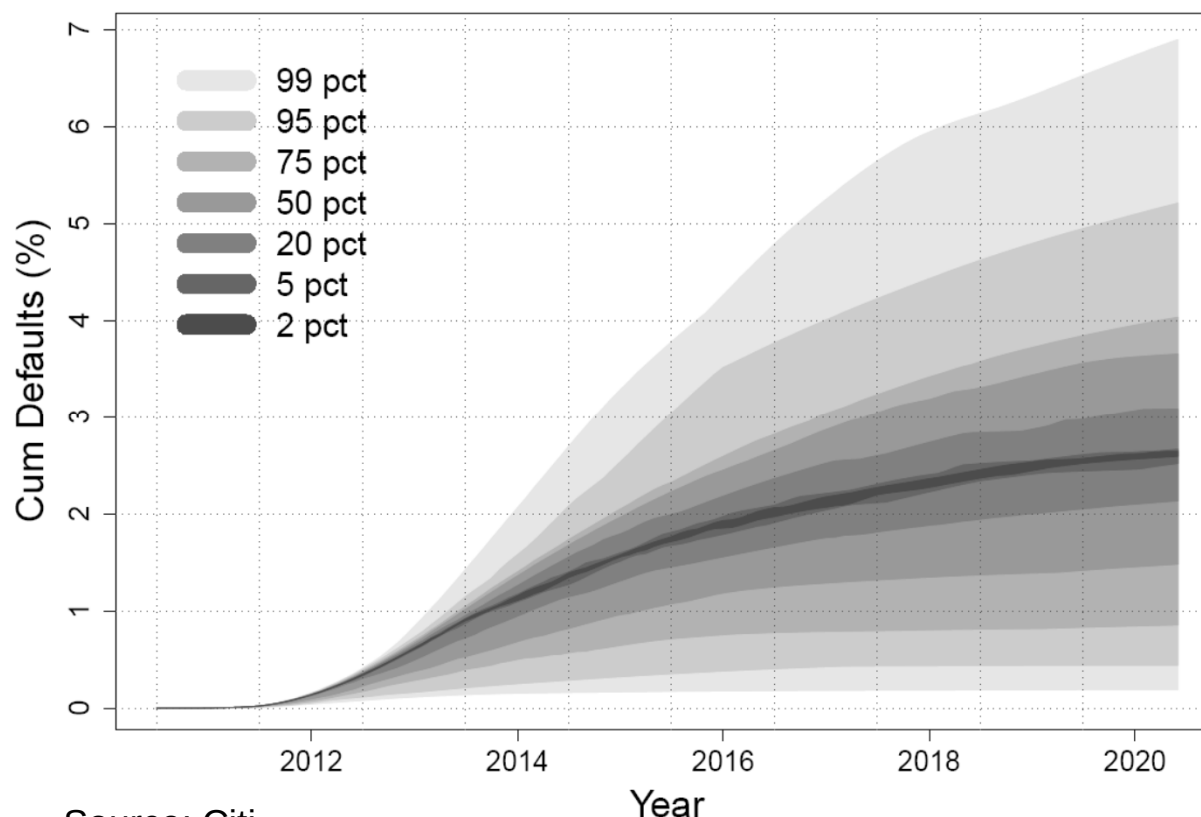


Source: Citi

Analysis of New Prime Loans: Envelope of Projected Cumulative Defaults

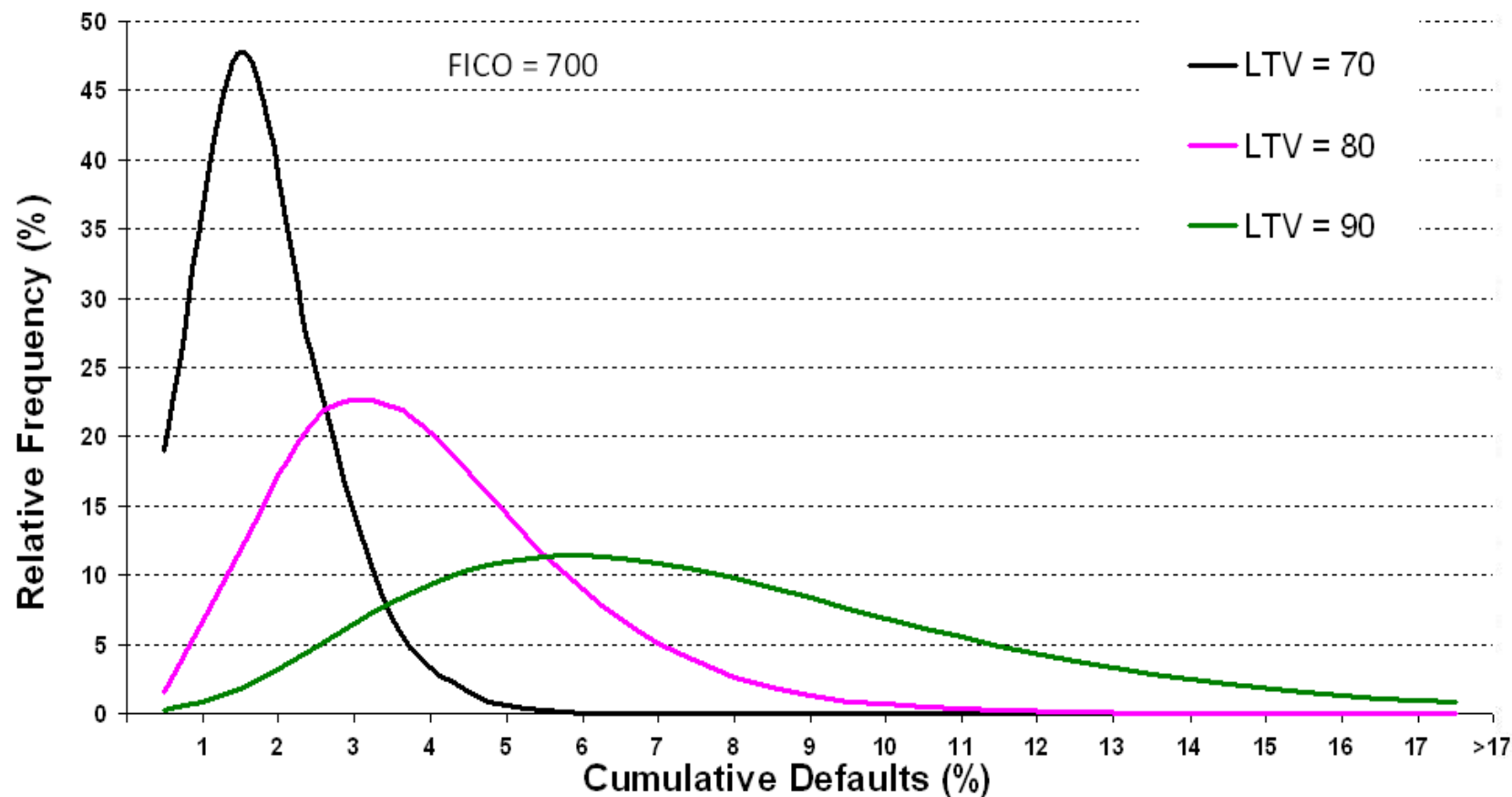


LTV=80%, FICO=750, DTI=30%



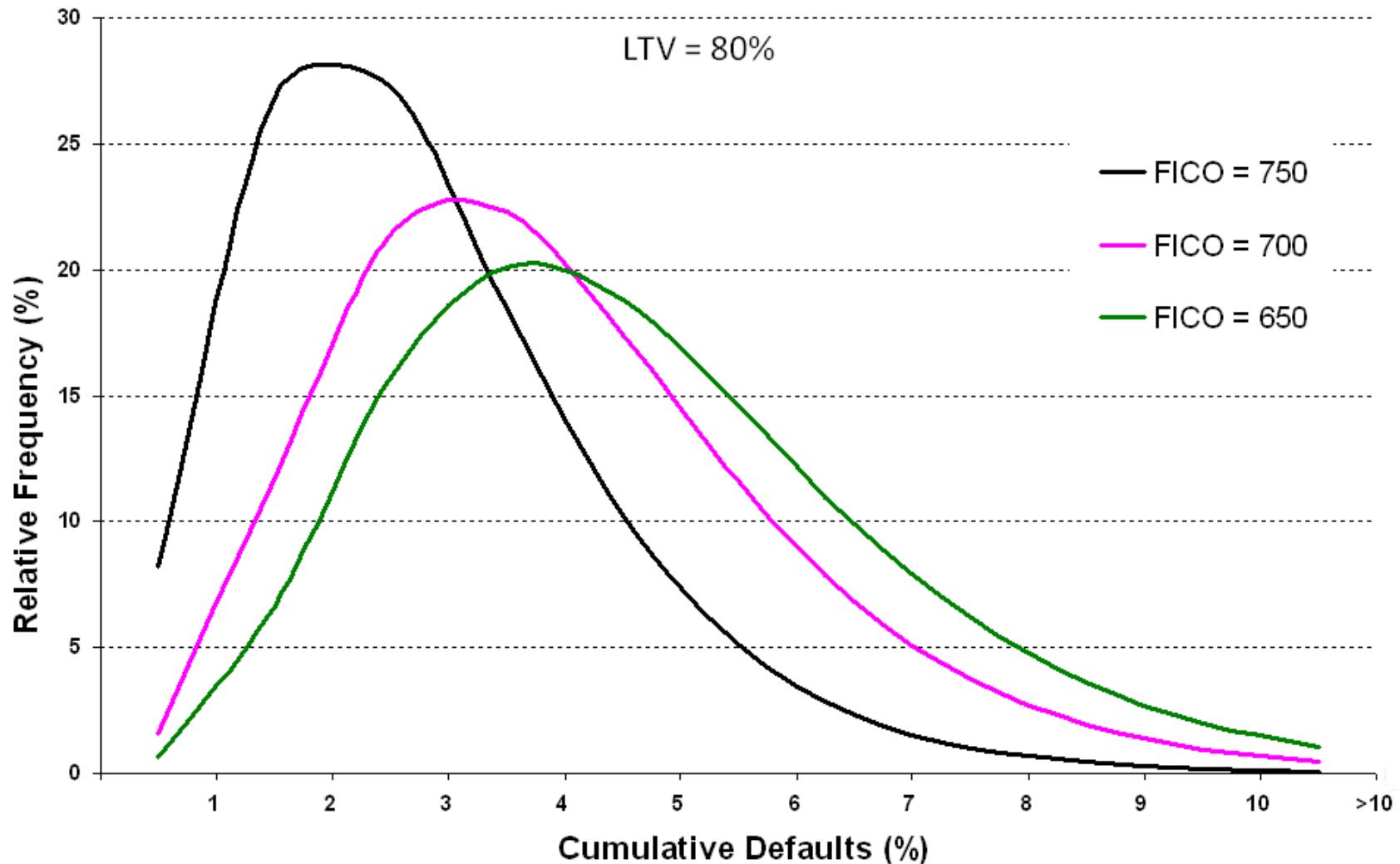
- Historical data on mortgage defaults can be used augment recent experience to model defaults in a post-bubble housing market;
- A summary of such data is give in *Evaluation of Credit-Sensitive Mortgage Securities*, Lakhbir Hayre, et al. Prudential Securities (1993)

Distribution of Projected Cumulative Defaults for New Loans with Varying Initial LTVs



Source: Citi

Distribution of Projected Cumulative Defaults for New Loans with Varying Credit Scores



Source: Citi

Credit-Adjusted OAS Analysis – Other Applications

HPA Duration

Given initial price PX and C-A OAS:

Shock Mean HPA Path +50bp:	C-A OAS to Price →	PX(HPA+50bps)
Shock Mean HPA Path –50bp:	C-A OAS to Price →	PX(HPA-50bps)

$$HPA\ Duration = 100 * [PX(HPA+50bps) - PX(HPA-50bps)] / PX$$

Separating Market and Credit Risk

R1= new interest rate level, H1= new HPA, PX = base price

$$\begin{aligned} \text{Risk} &= PX - PX(R1, H1) = PX - PX(R1) + PX(R1) - PX(R1, H1) \\ &= \text{Change due to Rates} + \text{Additional change due to HPA} \\ &= \text{Market Risk} + \text{Credit Risk} \end{aligned}$$

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